



NATIONAL ENERGY COUNCIL

EXECUTIVE REFERENCE DATA

Energy Management Towards
The National Energy Security

PREFACE

Indonesia's optimistic economic growth with an average growth rate of 7.6% per year during the period 2011 through 2012, has triggered the increase of domestic energy demand by an average of 4.4% per year. By 2030 the energy demand is estimated to be three times the demand level in 2010, while the increase in energy production in 2030 is estimated at twice that of the year 2010. This condition requires a good energy planning and management to ensure the supply of energy in the long term.

As to give a feature on the energy management condition in Indonesia, the National Energy Board deems necessary to publish a booklet entitled "Executive Reference Data: Energy Management towards National Energy Security". This first edition booklet in 2013, illustrates the energy management of Indonesia up to 2012. It presents information on the oil, gas, coal and electricity condition, the progress of 10,000 MW accelerated development of the electric power generation plants phase I and phase II, the development of new and renewable energy, CO₂ emissions from the energy users sector and of the Indonesia's energy position in the ASEAN region.

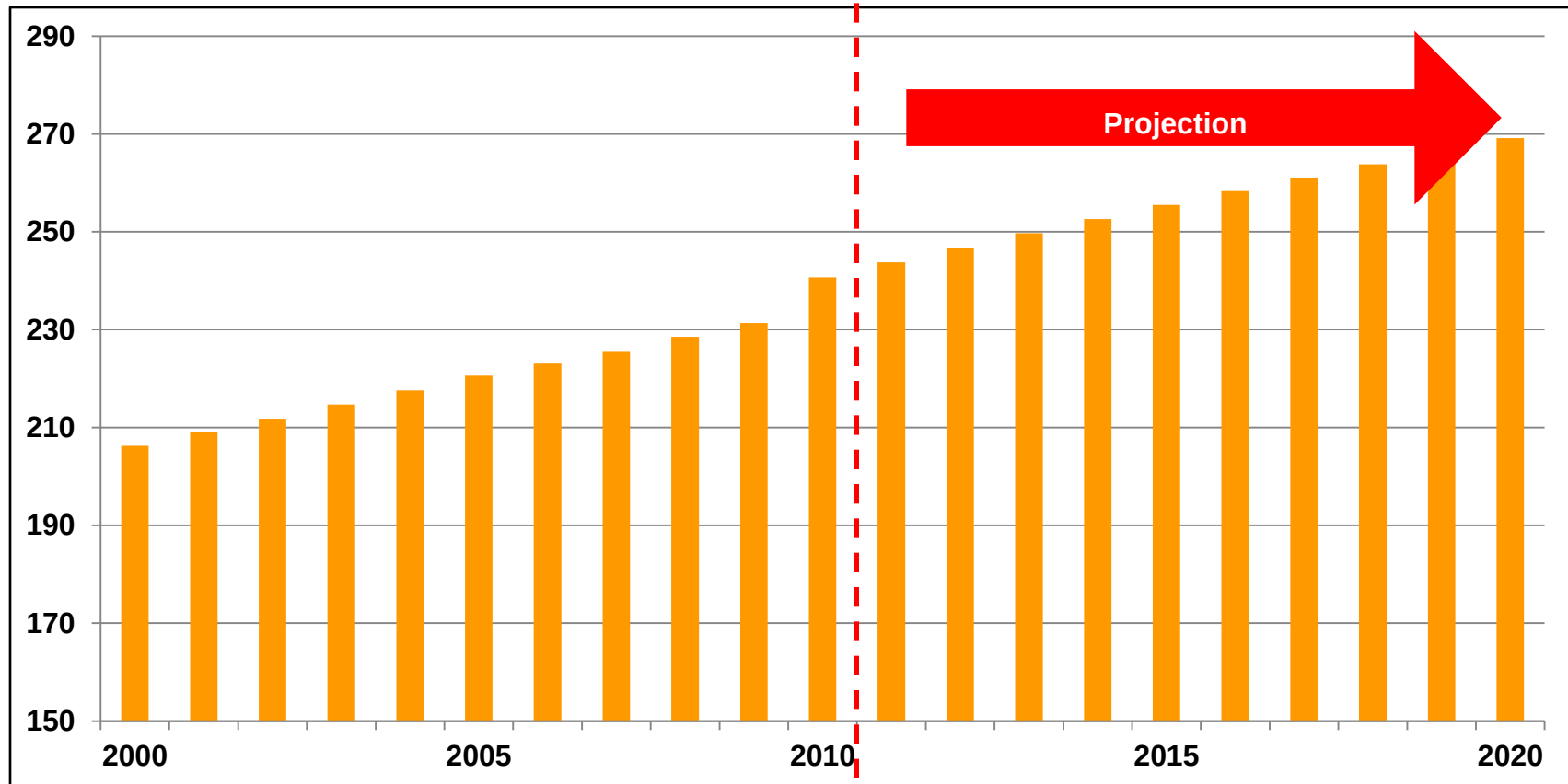
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SOCIAL ECONOMY

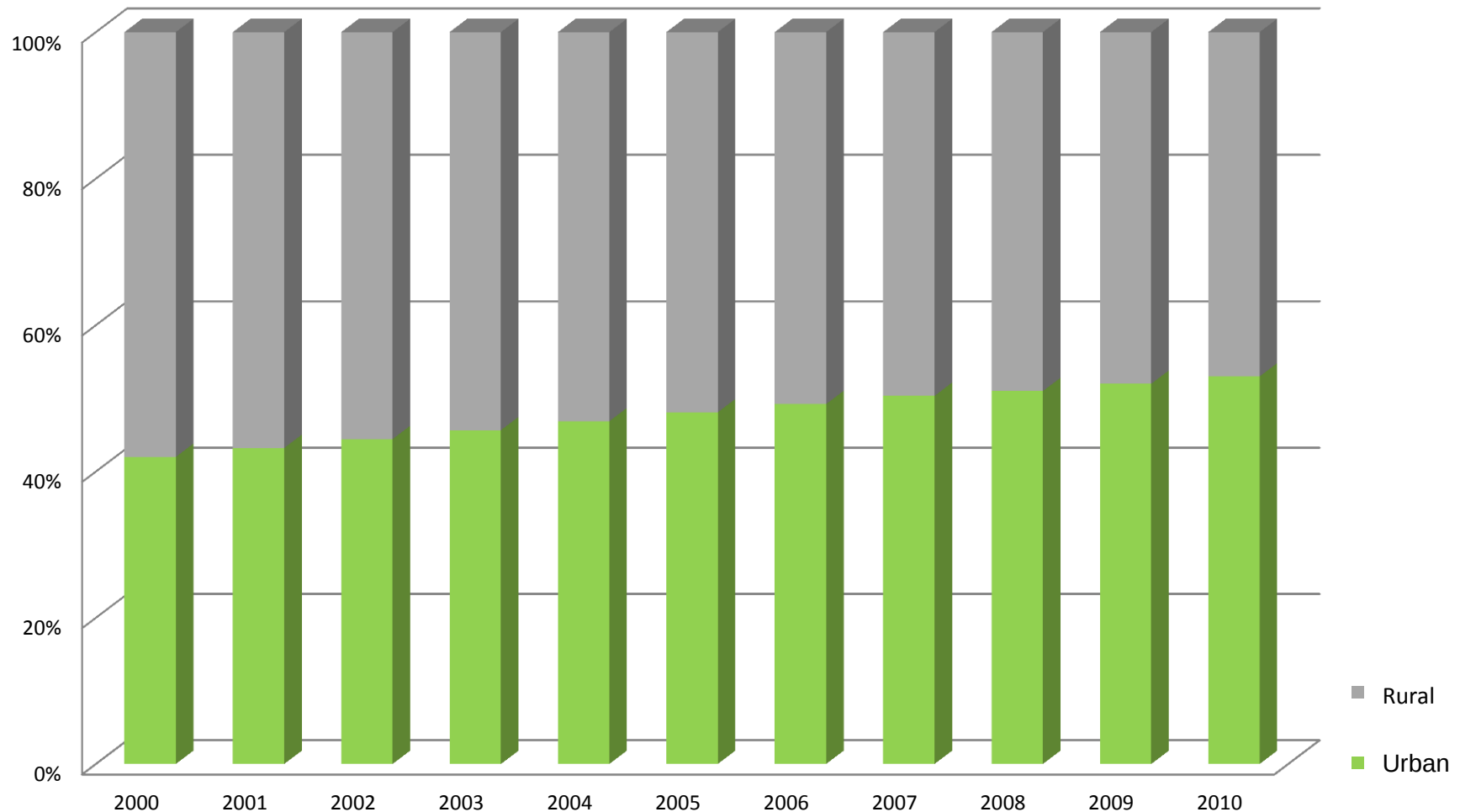
POPULATION AND PROJECTION

(Million people)



In 2010, Indonesia's population rose to 240 million people, with an annual average growth rate of 1.3% and would become 269 million people in 2020 (Source: Indonesian Central Bureau of Statistics)

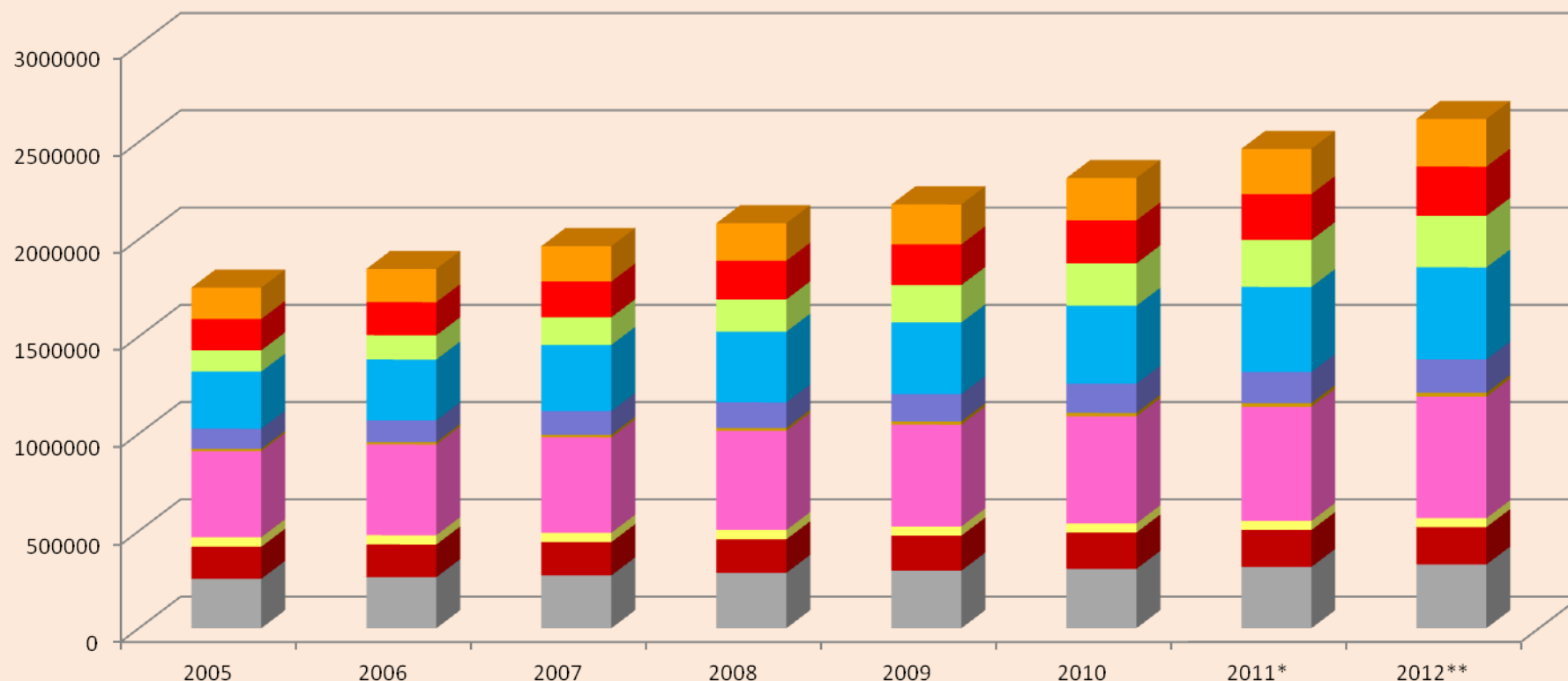
POPULATION IN RURAL AND URBAN AREAS



It is estimated that the population in the urban areas will increase in 2015, with the composition of town/city and village by 56% versus 44%. Changes of a village to become a town has caused the increase in number and density of population, economic activity is no longer relies on the agricultural sector, to the improvement of infrastructure. (Source: Central Bureau of Statistics Indonesia)

GDP BASED ON CONSTANT 2000 PRICES BY SECTOR

(Billion IDR)



- Agriculture, Livestock, Forestry and Fisheries
- Mining and Quarrying
- Oil and Gas Industry
- Non Oil and Gas Industry
- Electricity, Gas & Water
- Construction
- Trade, Hotel & Restoran
- Cargo dan Communication
- Finance, Real Estate & Company Service
- Services

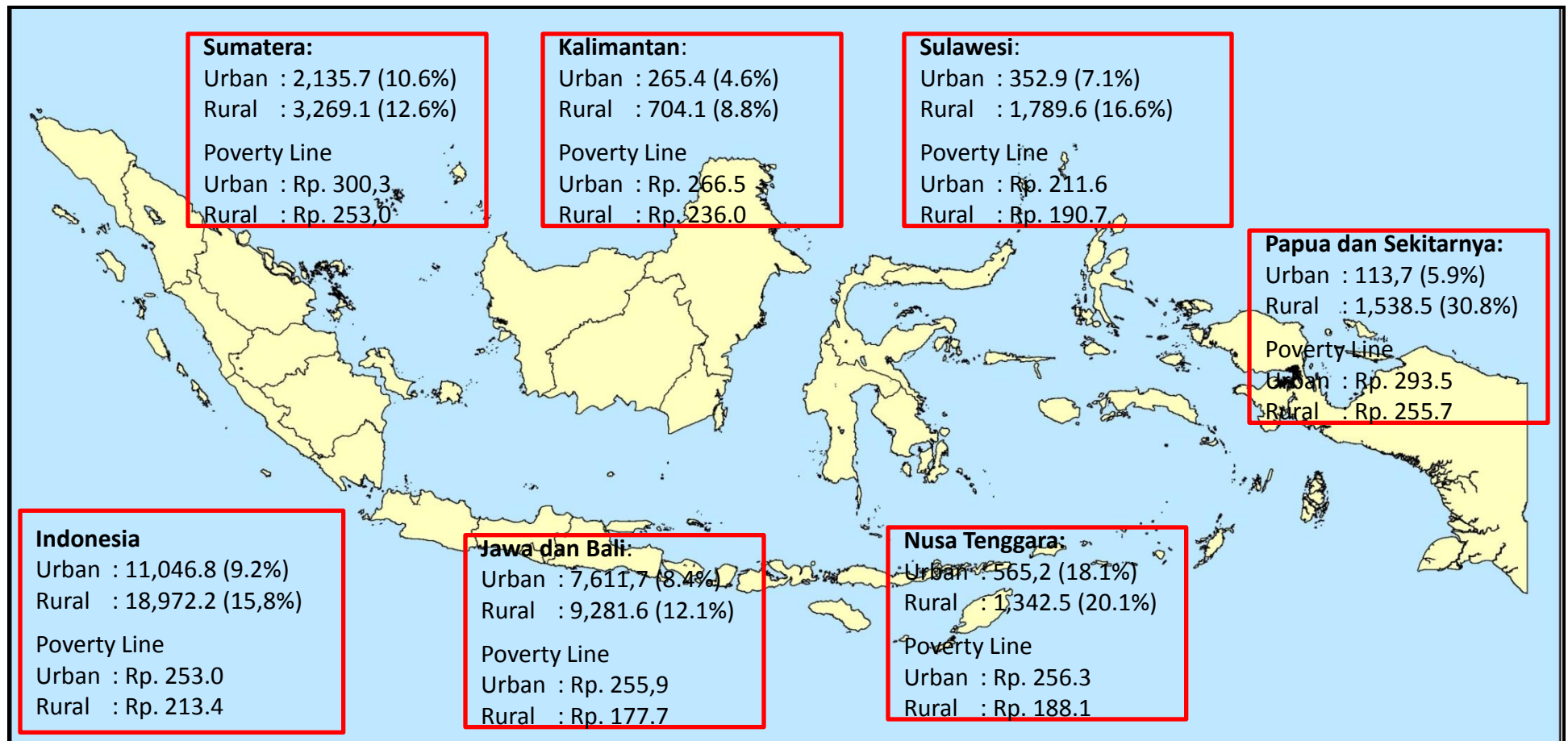
* Temporary Value

** Very Temporary Value

Indonesia's GDP increased more than 6% per year despite the world crisis. Based on BPS data, the structure of GDP in 2012 was dominated by the manufacturing sector reached 24%, the trade, hotels and restaurants for 18%, agriculture 13%.

POOR PEOPLE AND POVERTY BORDER LINE 2011

(.000)



In 2011, Urban population in Indonesia was 11.04 million people, of which 9.2% of the total were the poor. While the number of people left rural areas was 18.97 million people, of which 15.8% of the total were categorised as the poor. The urban poverty line level was 253 thousand rupiah, and in the rural area was 213 thousand rupiah.

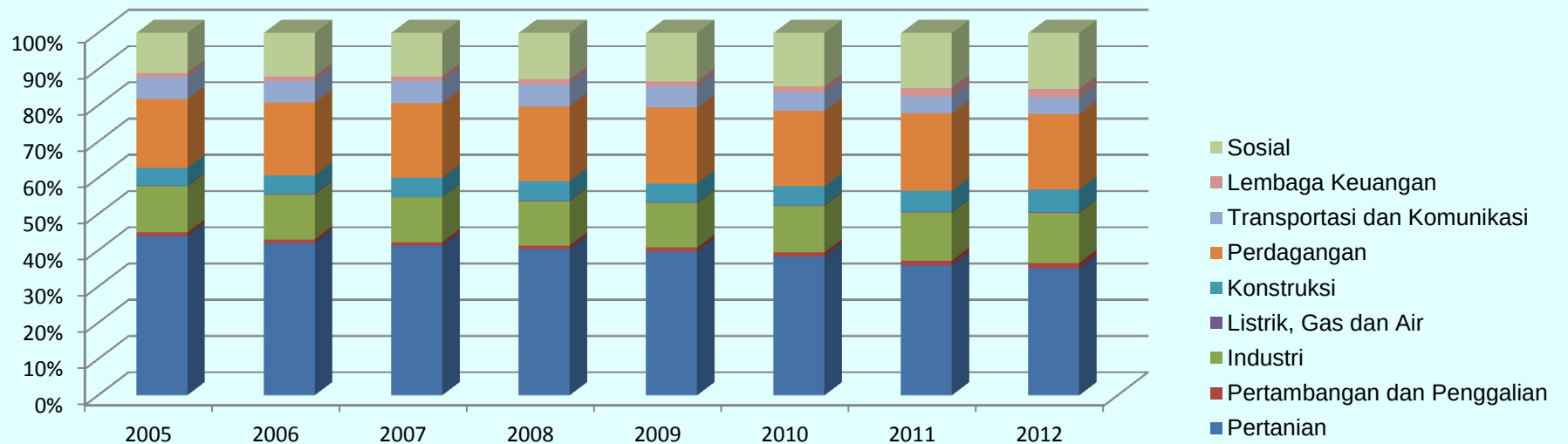
WORK AND LABOUR

Work force and the unemployment

(Million People)



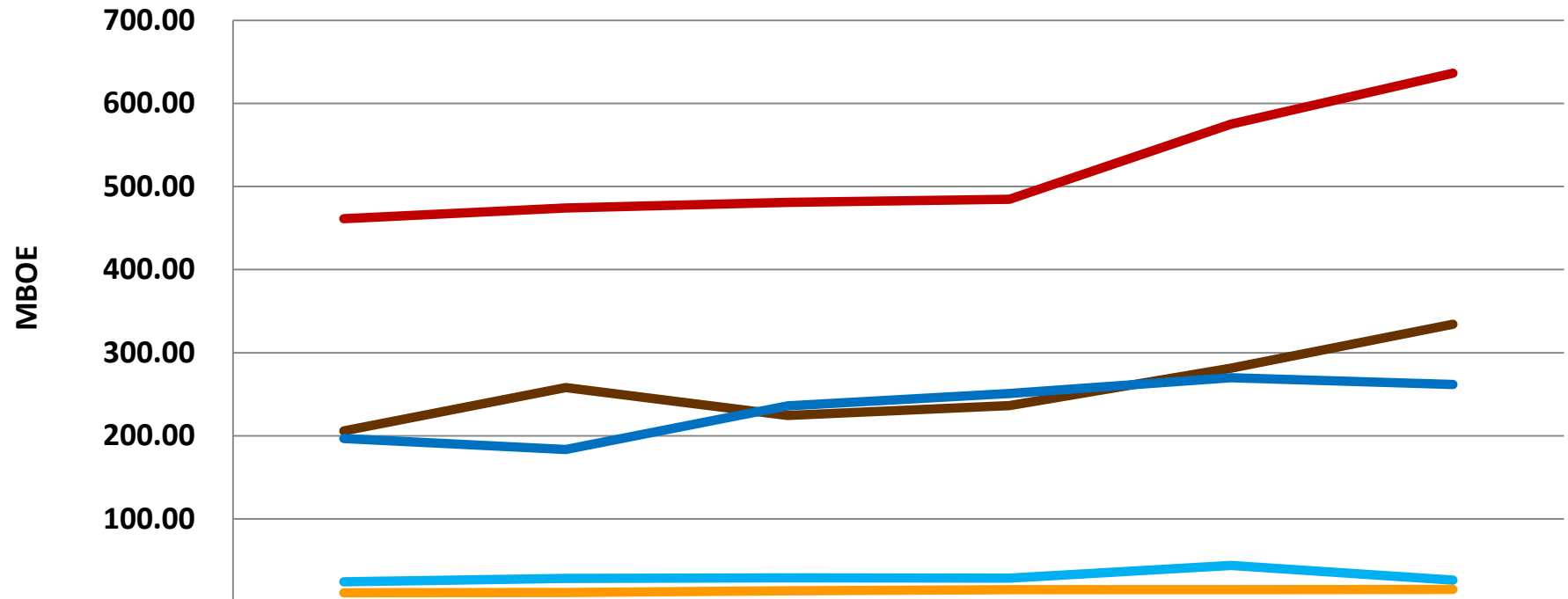
Percentage of Main Job-Fields



ENERGY INDICATOR

PRIMARY ENERGY SUPPLY

(MBOE)

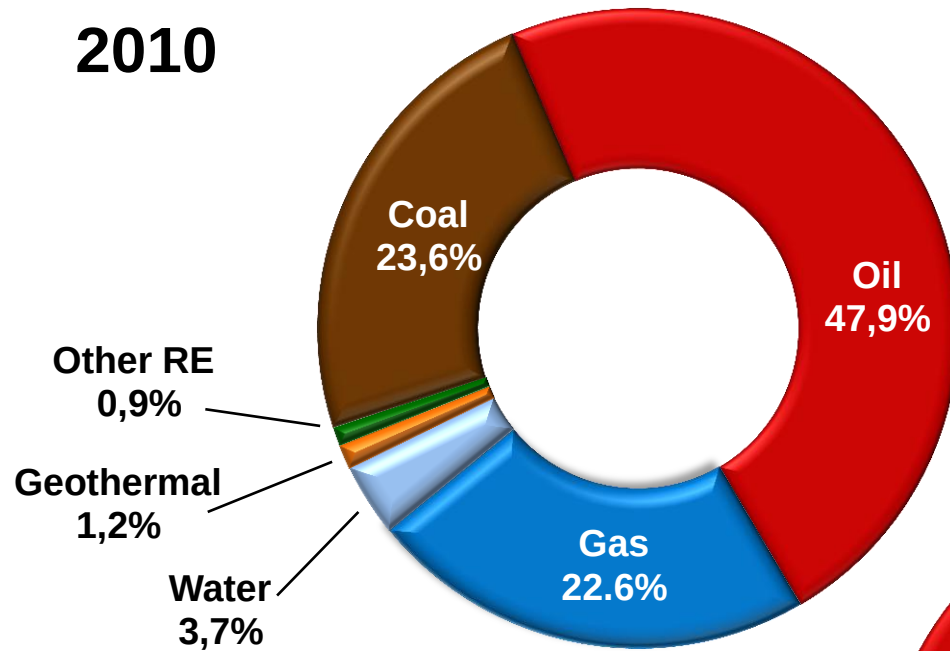


	2006	2007	2008	2009	2010	2011
— Batubara	205.78	258.17	224.59	236.44	281.40	334.14
— Minyak dan Produk	461.35	474.03	480.90	484.73	575.25	636.52
— Gas dan Produk	196.60	183.62	236.05	251.04	269.94	261.71
— Tenaga Air	24.26	28.45	29.06	28.70	44.00	26.37
— Panas Bumi	11.18	11.42	13.42	14.97	14.68	15.09

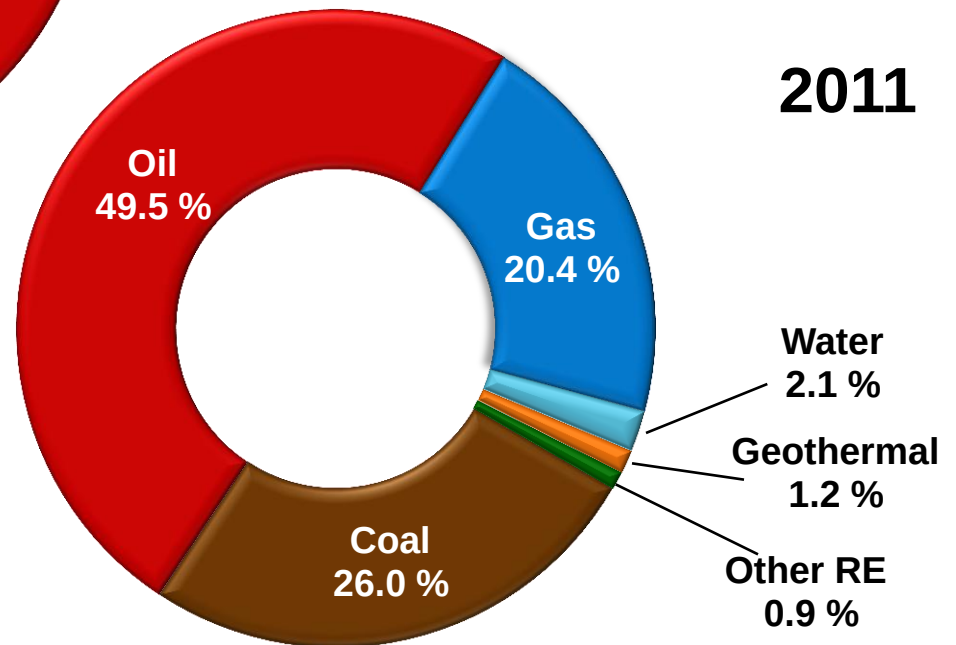
Note : Exclude Biomass

ENERGY MIX

2010

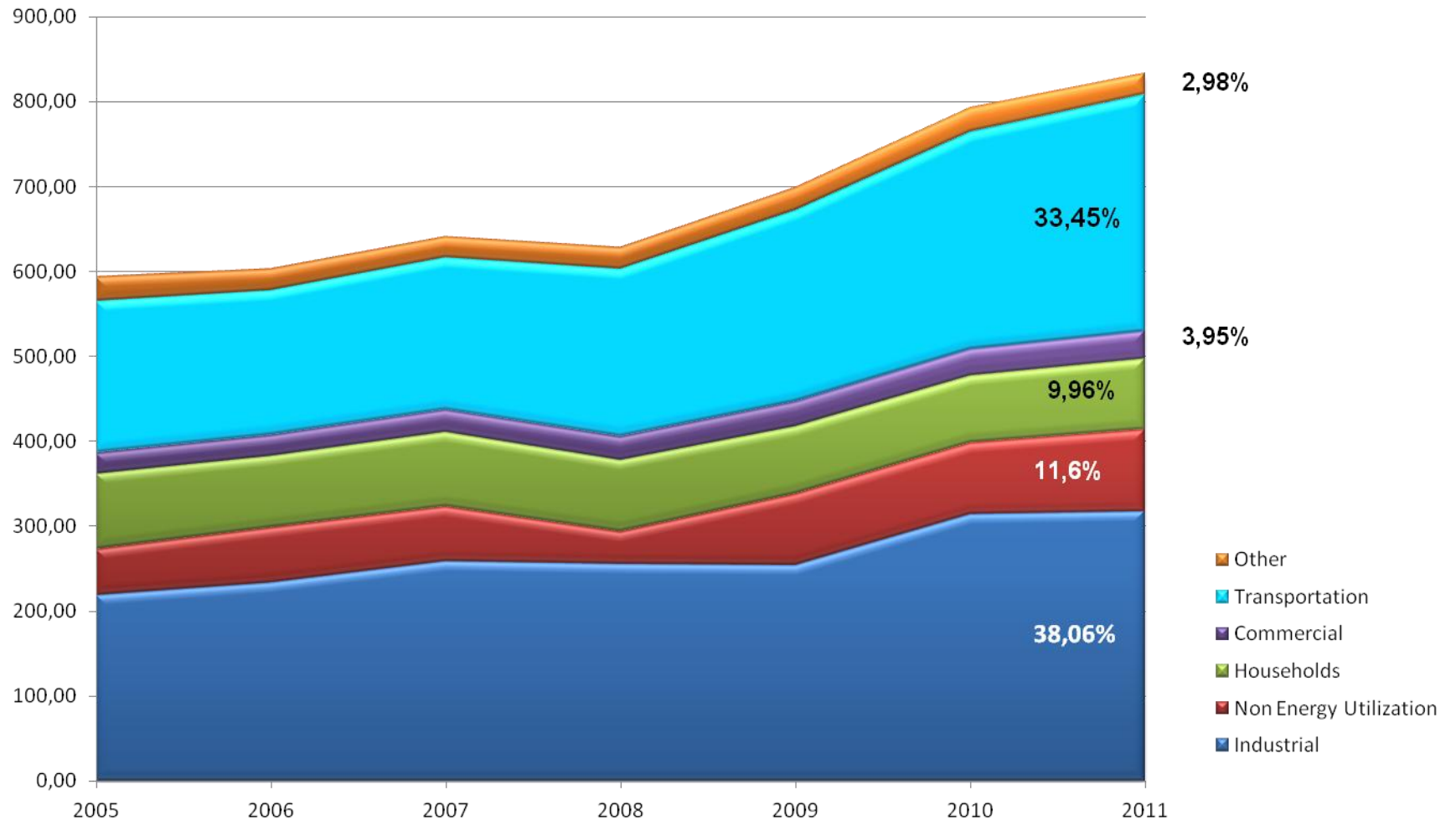


2011



FINAL ENERGY CONSUMPTION BY SECTOR

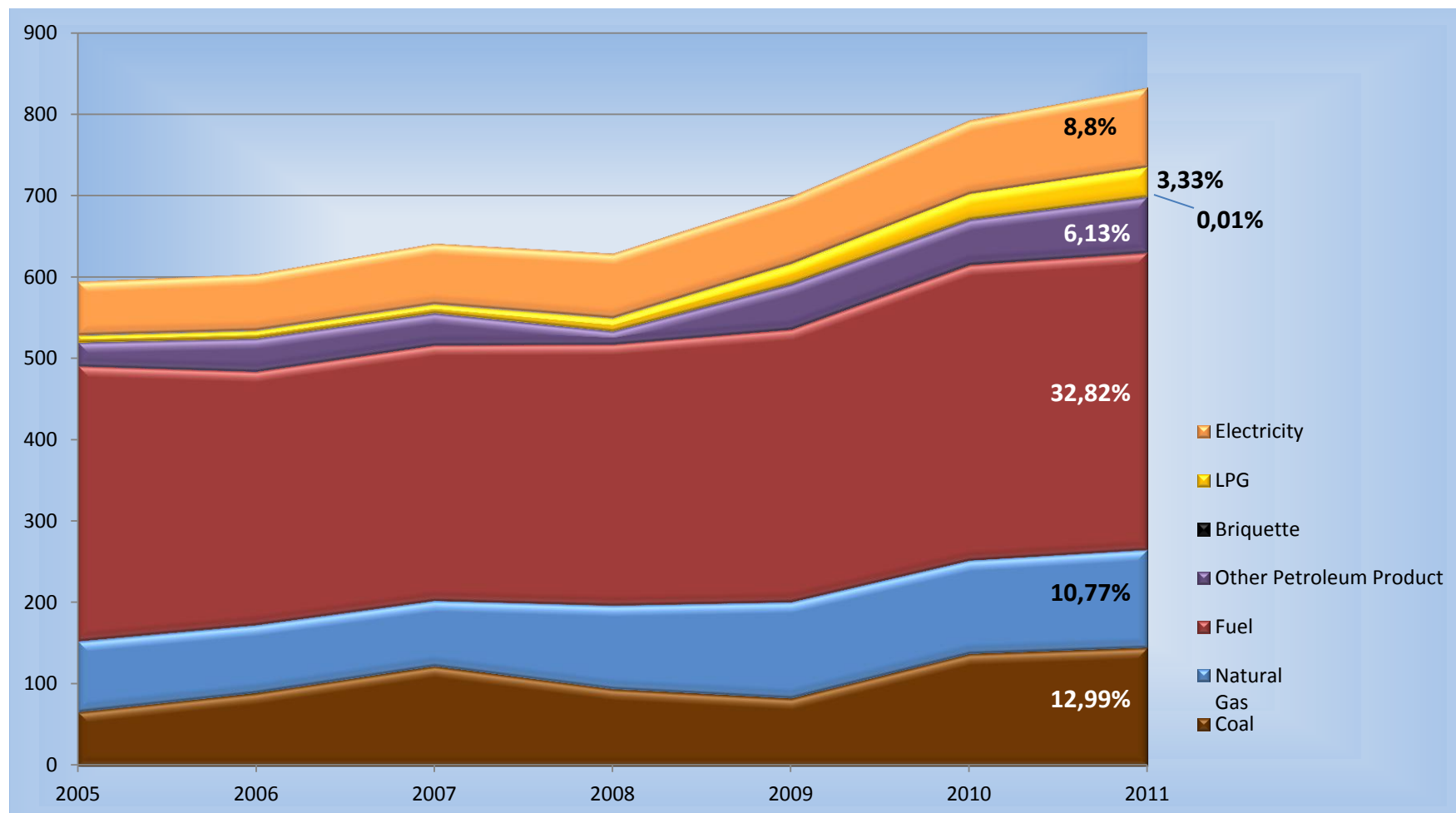
(MBOE)



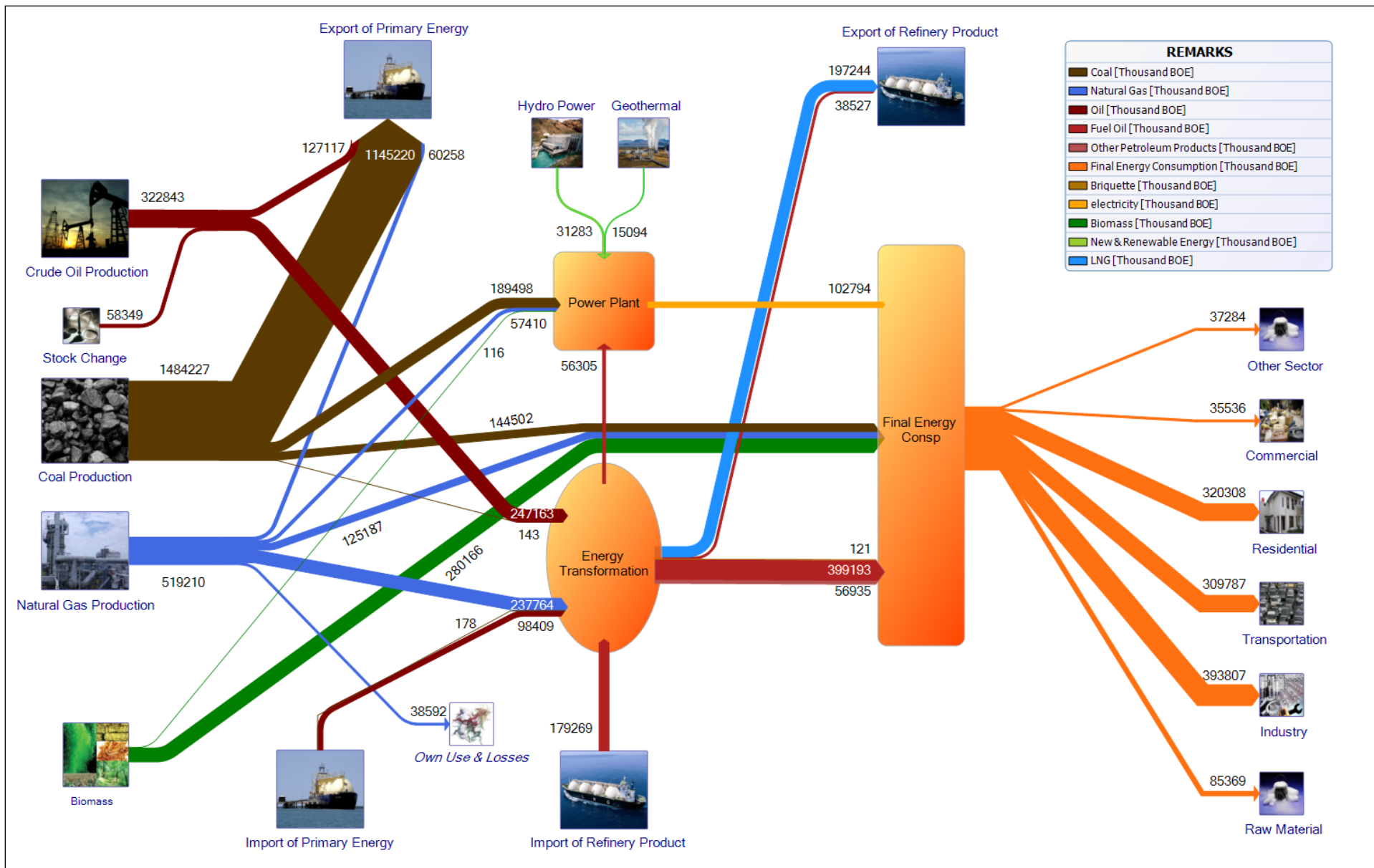
Exclude Biomass

FINAL ENERGY CONSUMPTION

(MBOE)

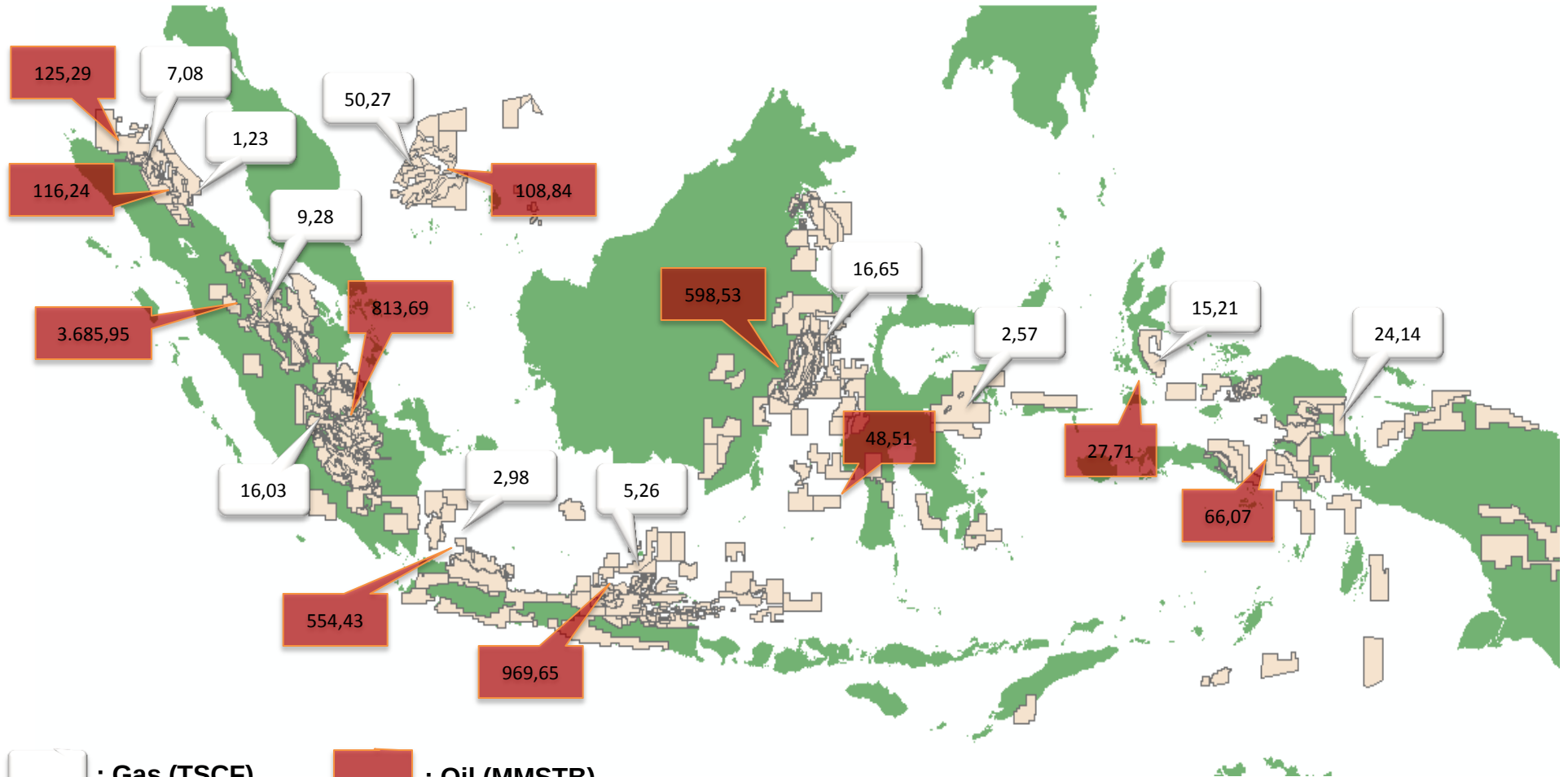


ENERGY FLOW 2011



OIL AND GAS

OIL AND GAS RESERVES



 : Gas (TSCF)

Proven = 103.35 TSCF
Potential = 47.35 TSCF
Total = 150.70 TSCF

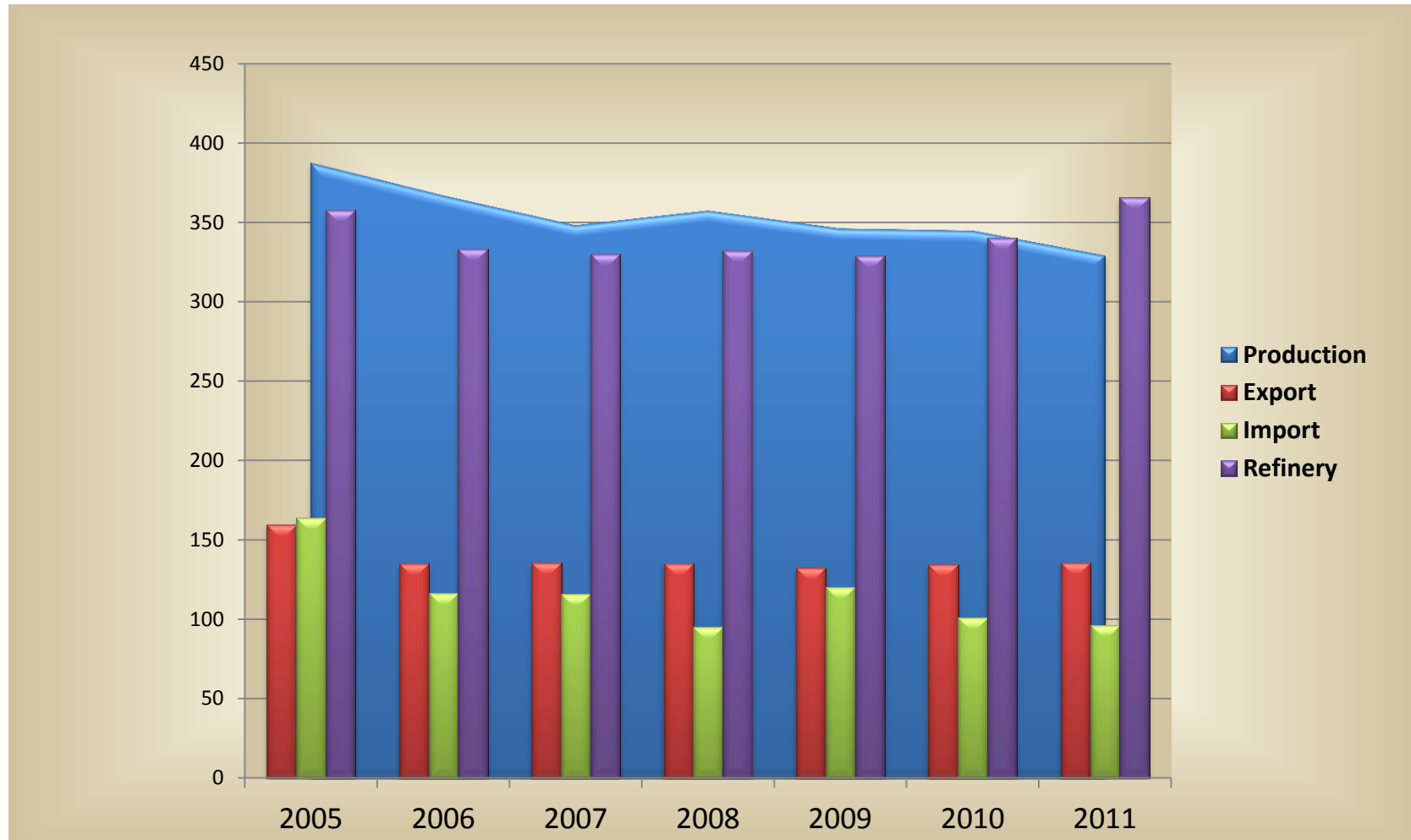
 : Oil (MMSTB)

Proven = 3,741.33 MMSTB
Potential = 3,666.91 MMSTB
Total = 7,408.24 MMSTB

Source : Ditjen Migas 2011,

OIL SUPPLY

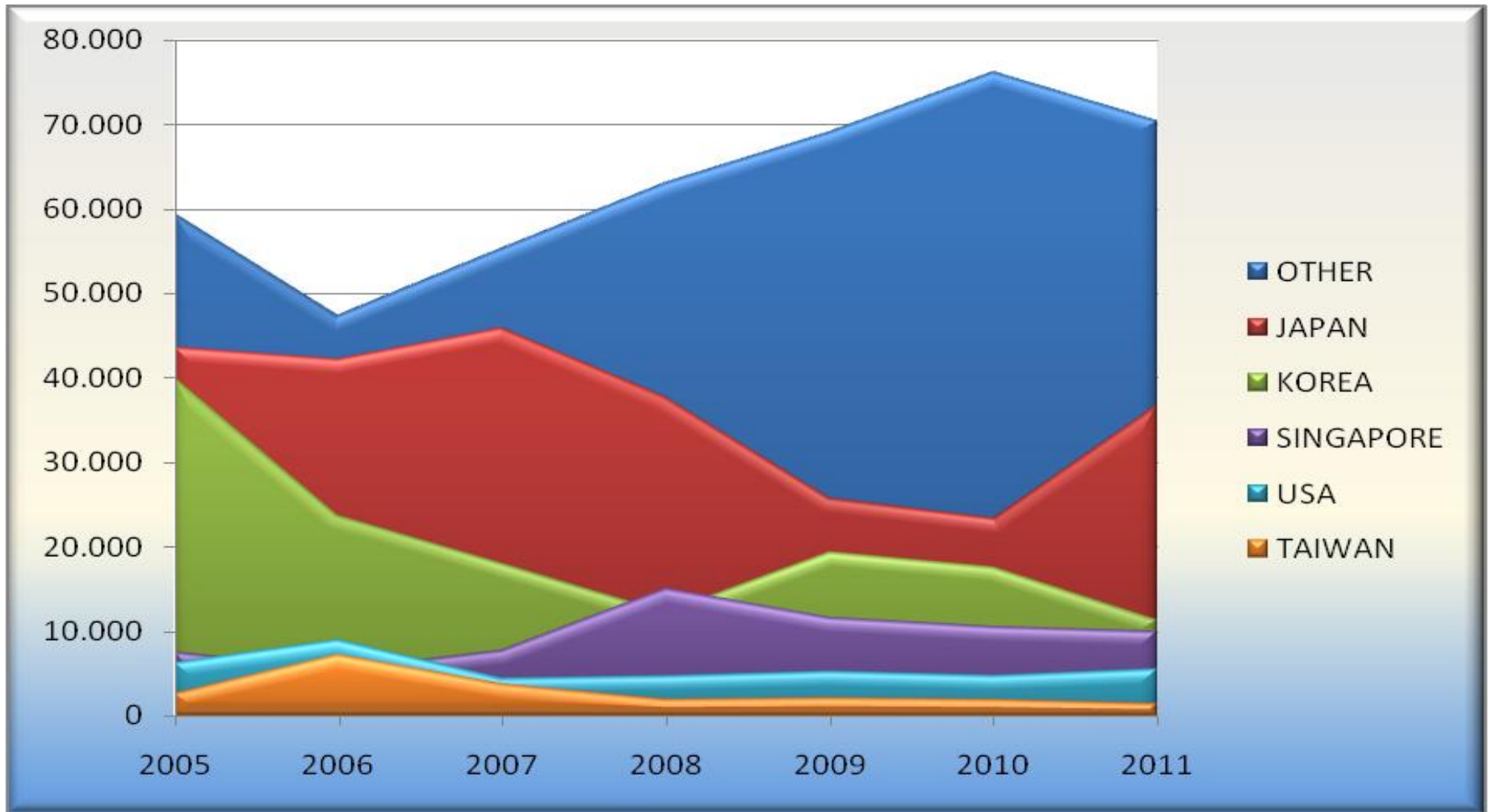
(Thousand Bbl)



*Note : discrepancy between production, export, import and domestic demand due to the existing stock
the role of imported crude in 2011 was amounted to 26,25 % from the total that goes to the oil refinery*

CRUDE OIL EXPORTS

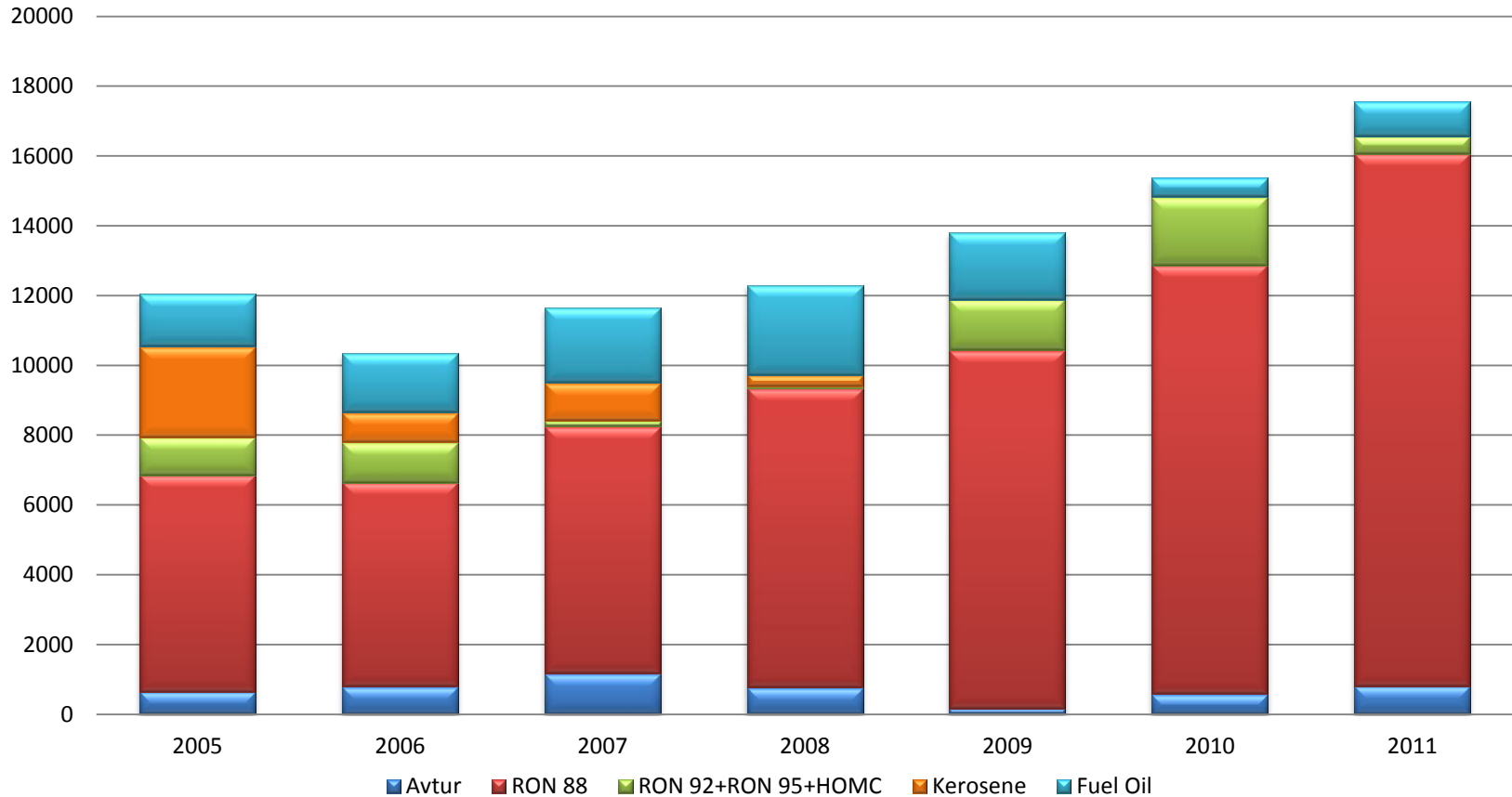
(Thousand Bbl)



Source : MEMR

FUEL IMPORTS

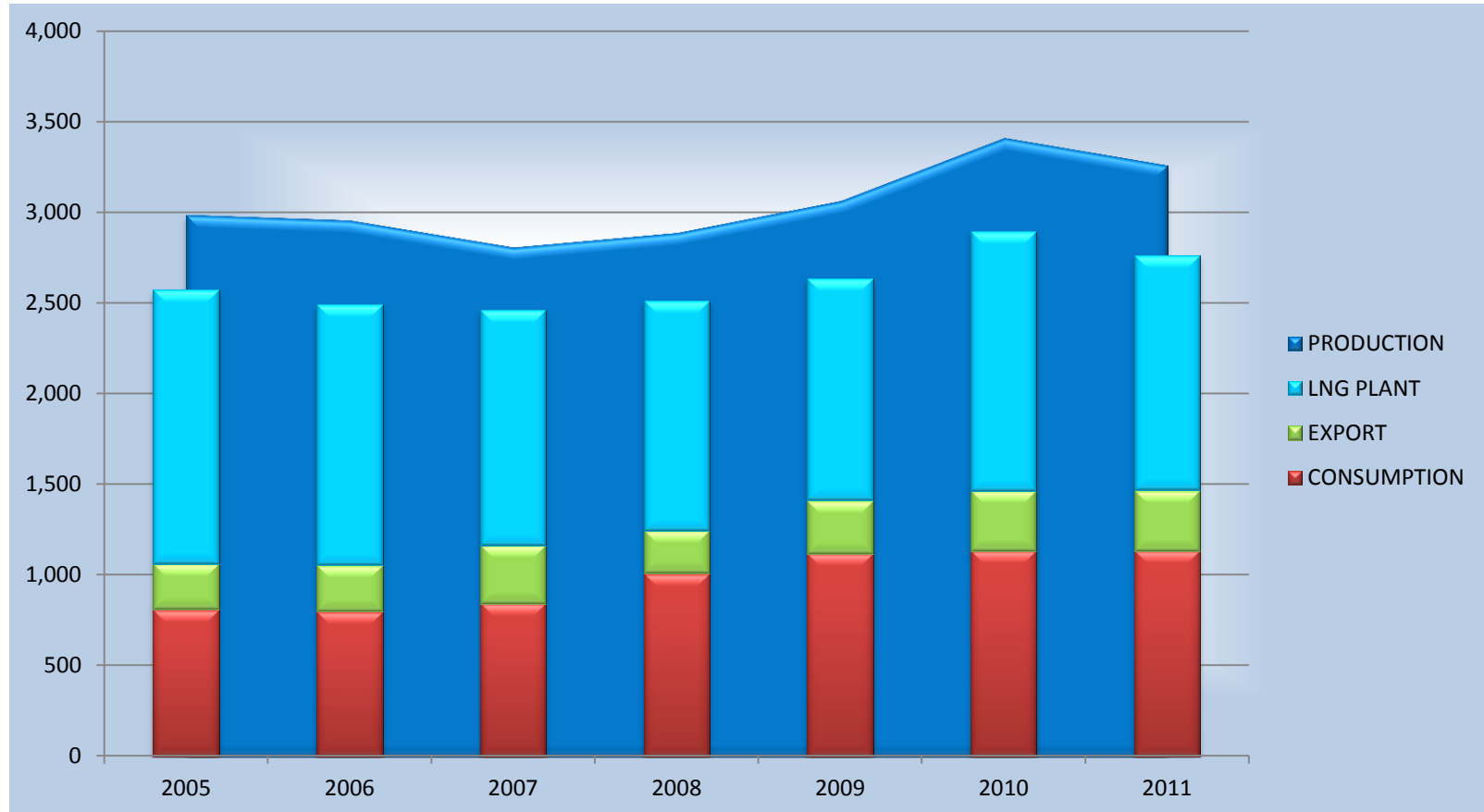
(Thousand KL)



Note : - The role of fuel imports in 2011 amounted to 0.04% of the total domestic fuel oil consumption
- The value of imports of RON 88 (premium) increased along with the increase in domestic demand
- Kerosene imports declined and disappeared in 2009 due to LPG conversion policy

GAS SUPPLY

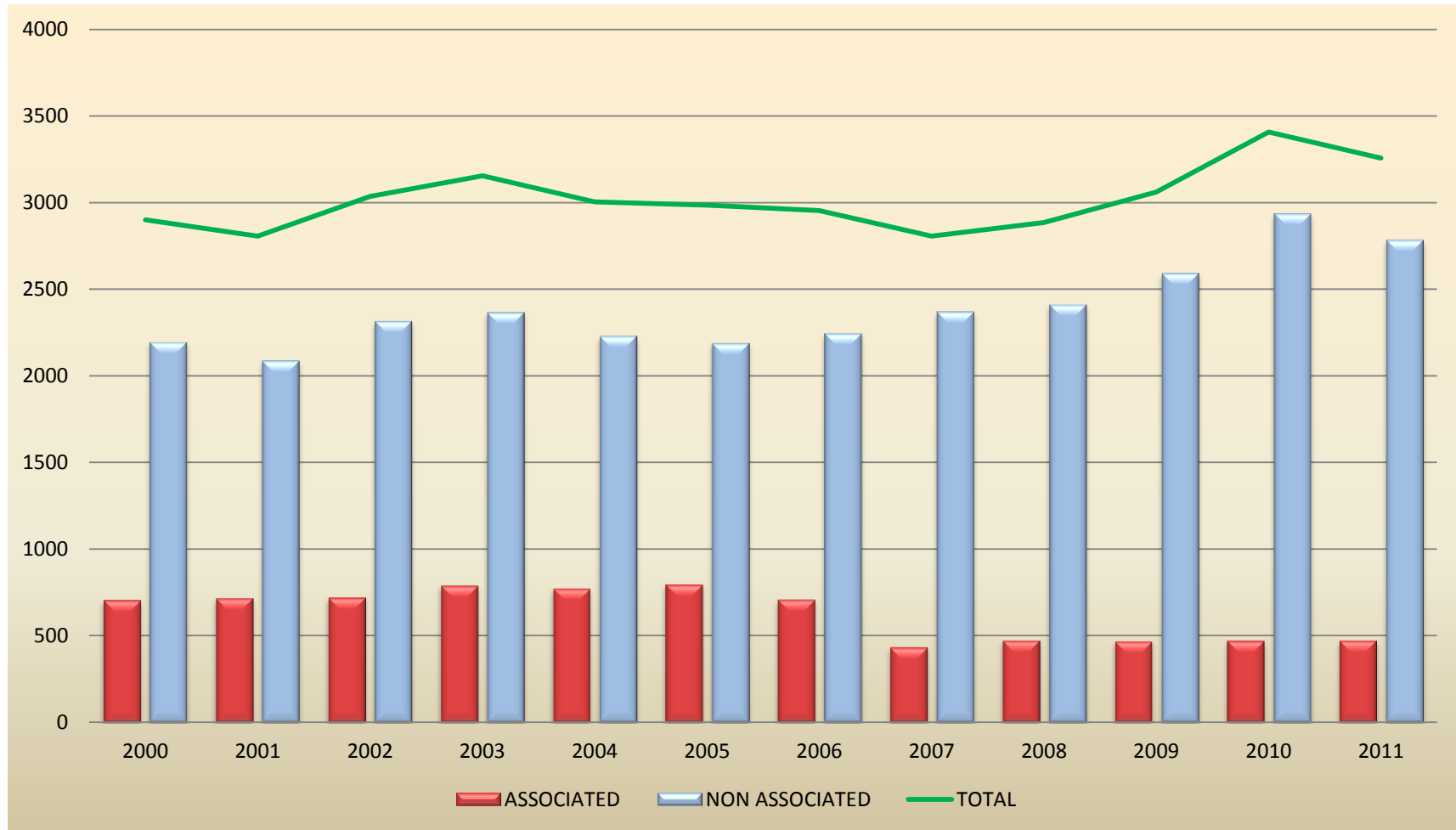
(Thousand MMSCF)



Note : - export means export of gas through pipeline
- products of LNG refinery allocated for exports
- discrepancy between the production and the total usage of gas due to gas reinjection, own use and flare

GAS PRODUCTION

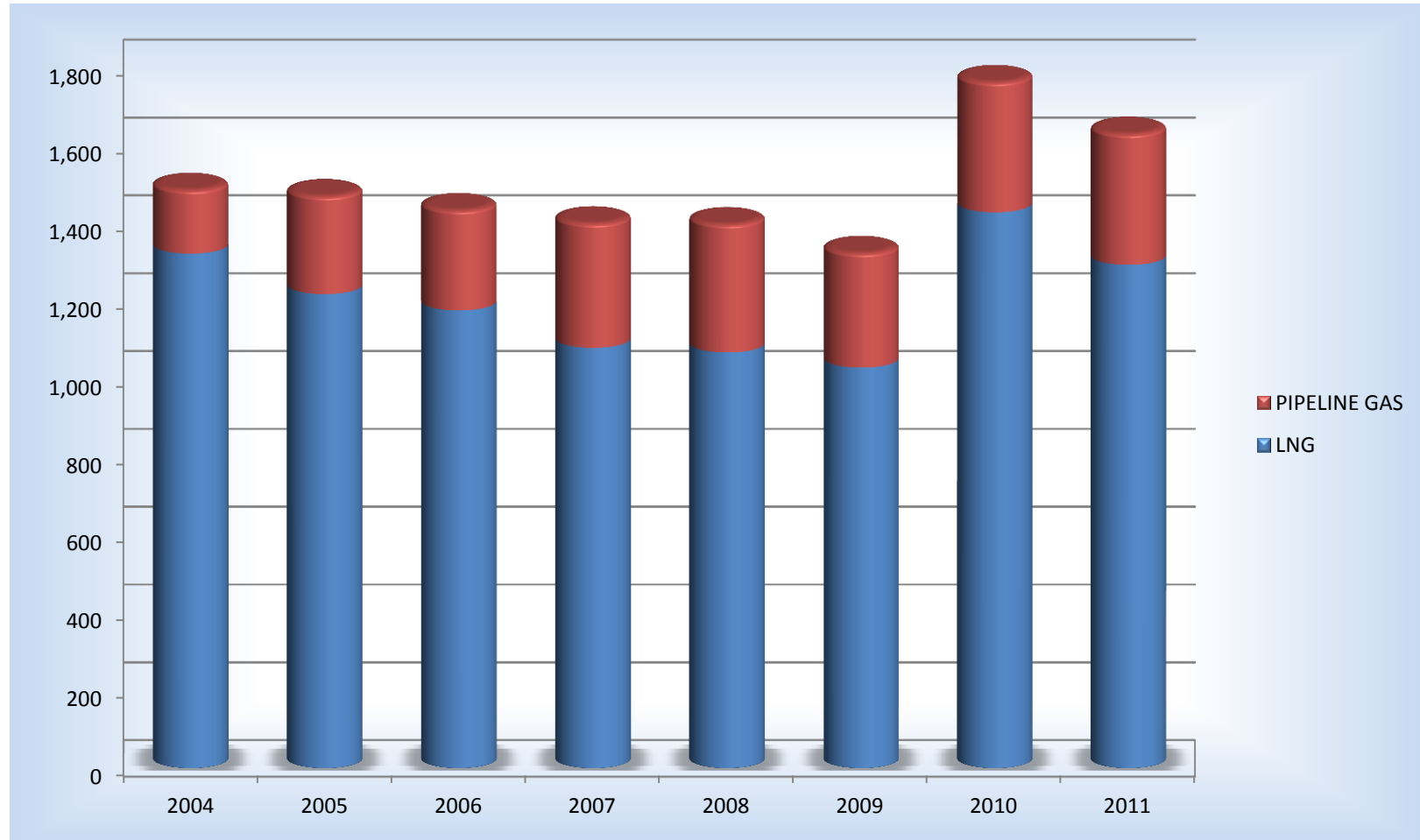
(Thousand MMSCF)



Note : associated defined as a natural gas found in association with oil, either dissolved in the oil or as a cap of free gas above the oil.

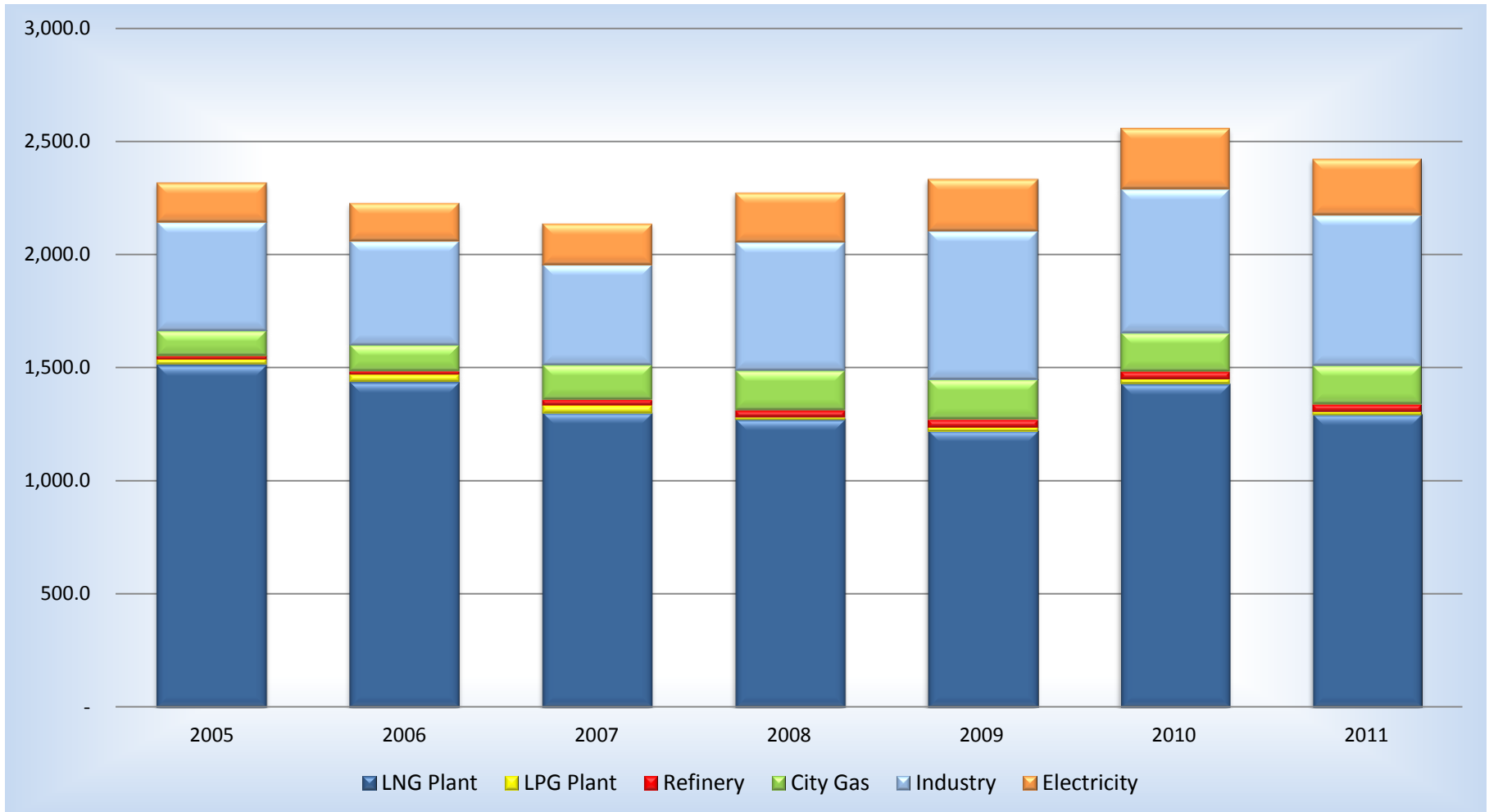
NATURAL GAS EXPORTS

(Thousand MMSCF)



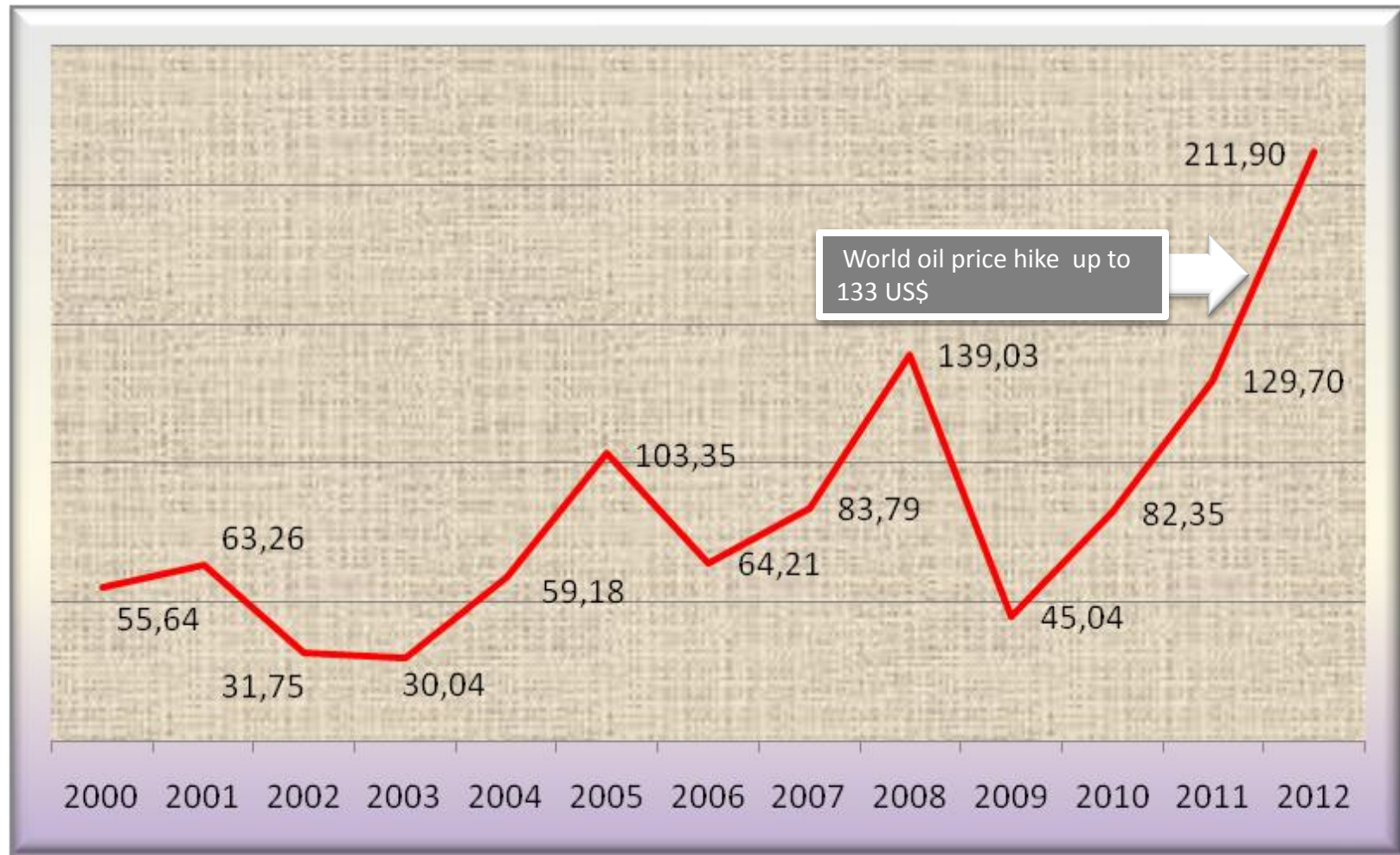
NATURAL GAS CONSUMPTION

(Thousand MMSCF)



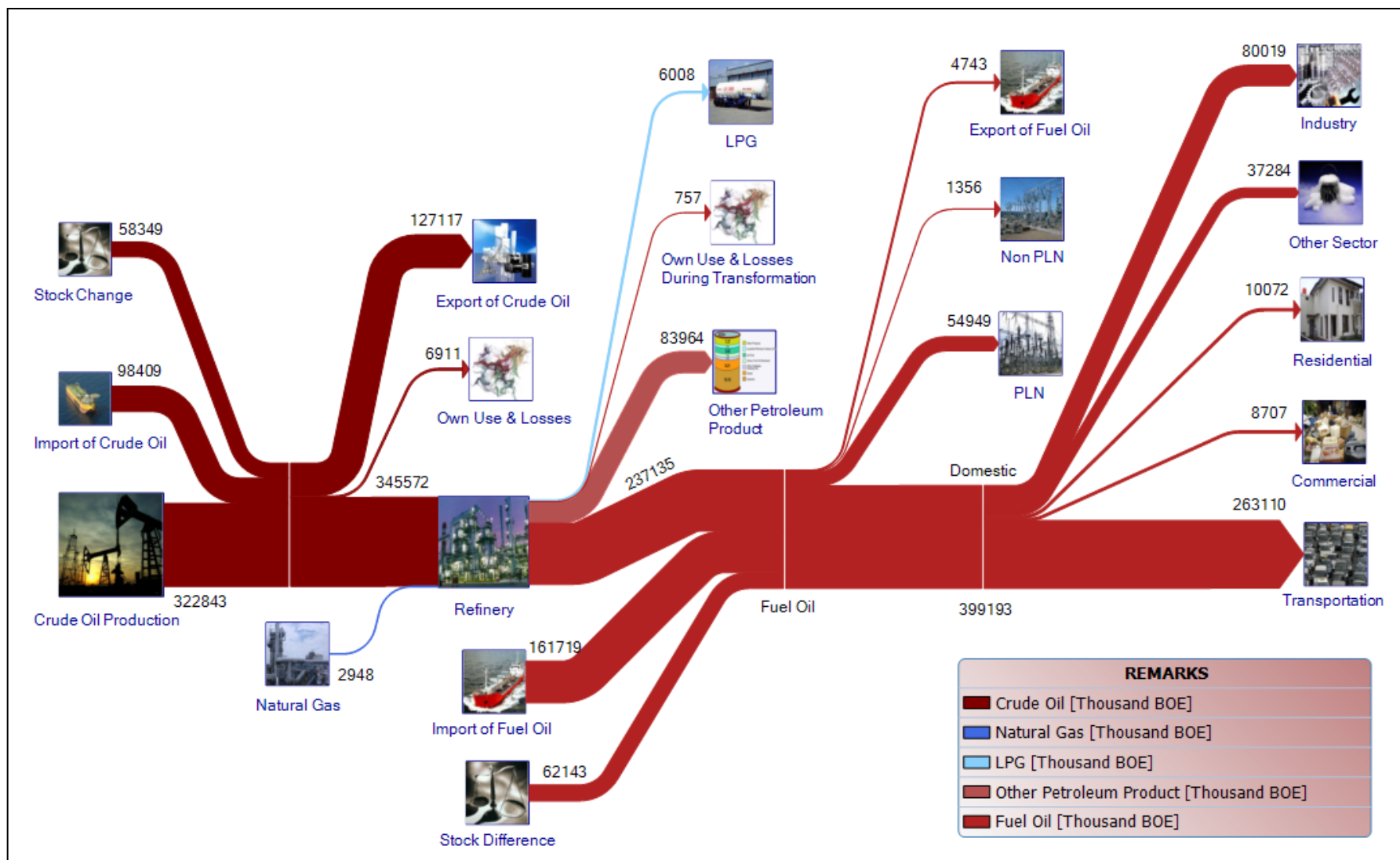
FUEL OIL SUBSIDY DEVELOPMENT

(Trillion IDR)

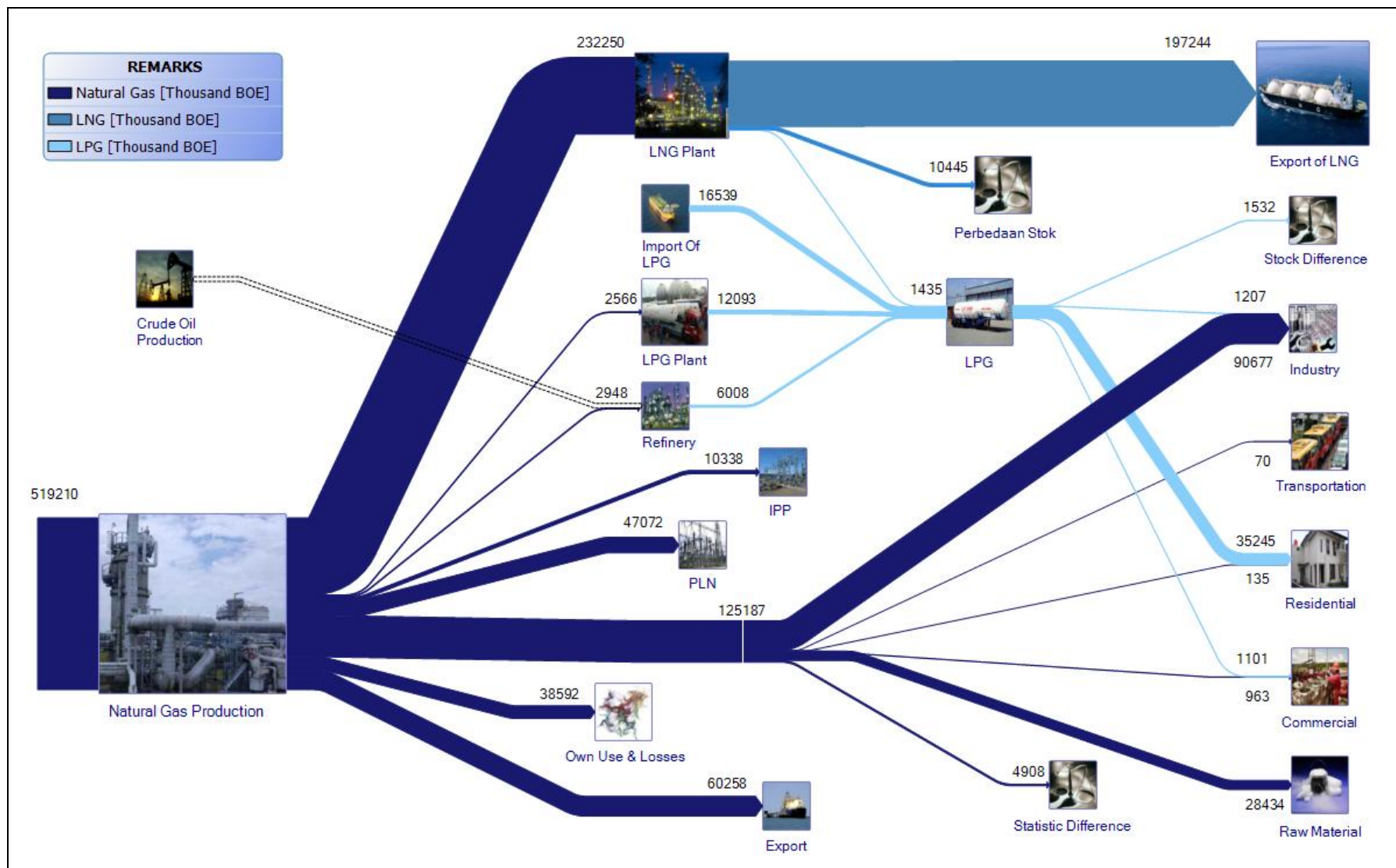


Note : rise in oil prices due to geopolitical crisis in the Middle East and Libya

OIL FLOW IN 2011

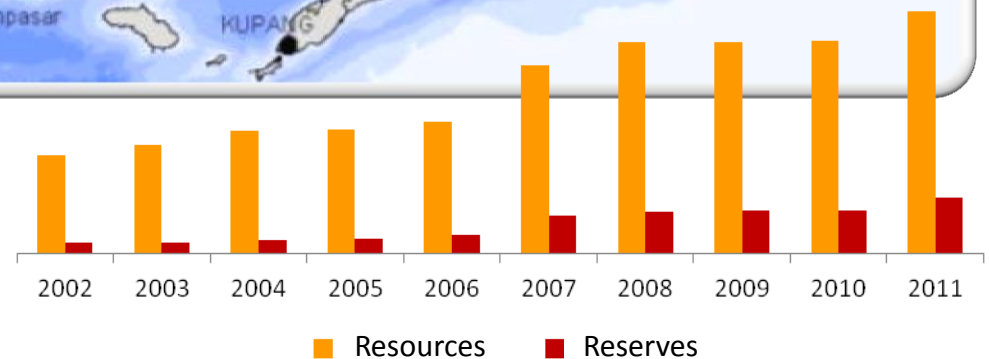
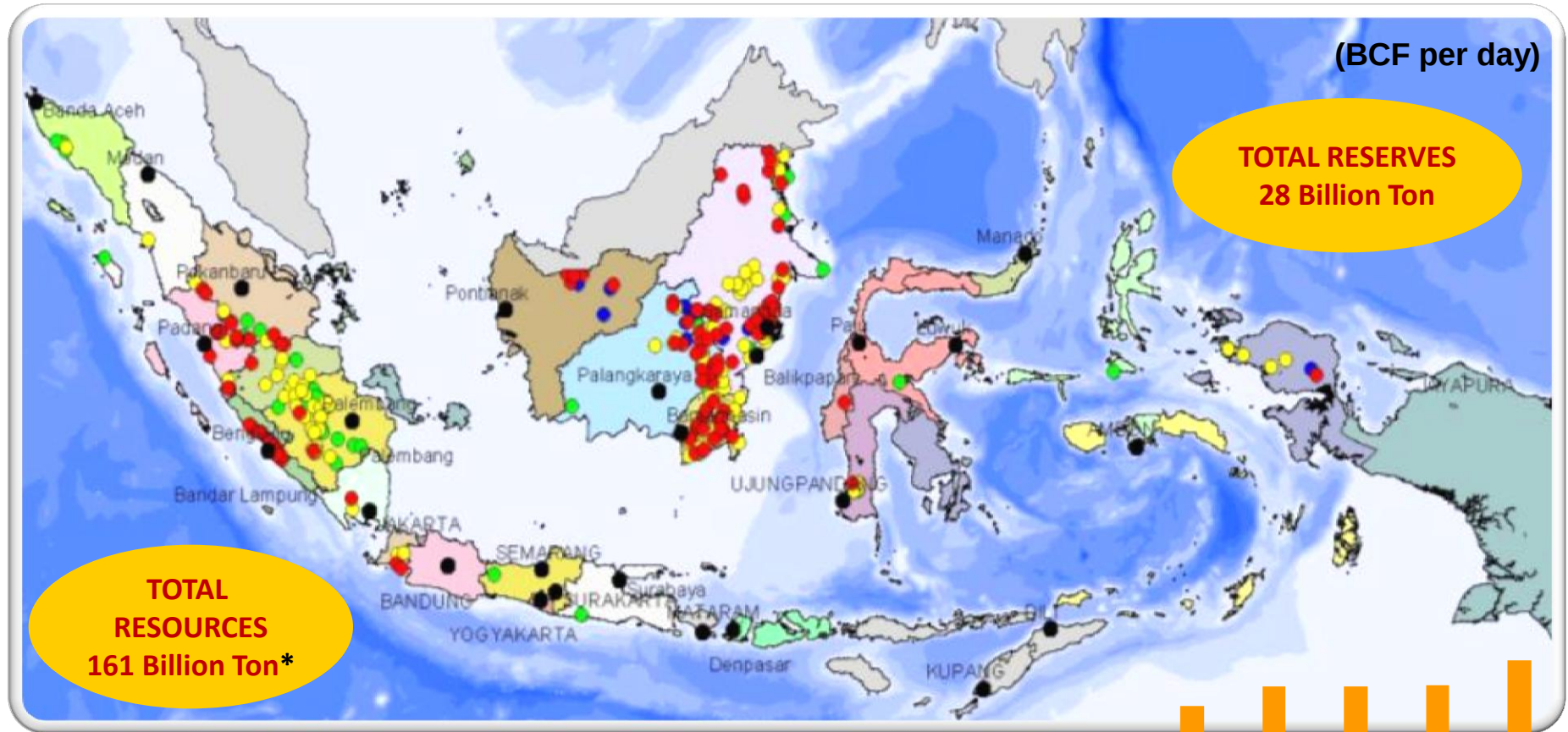


GAS FLOW IN 2011

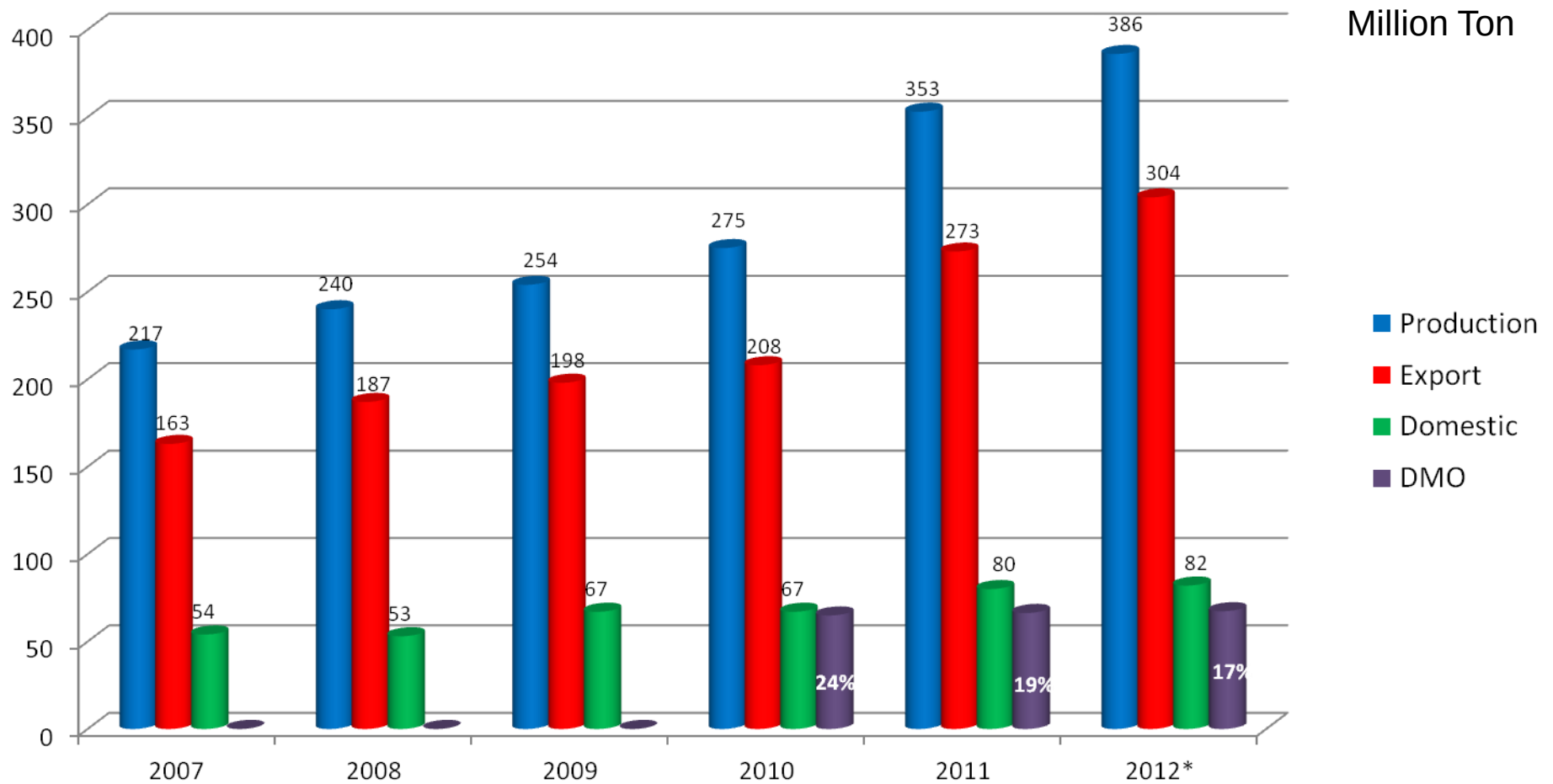


COAL

RESOURCES AND RESERVES 2011



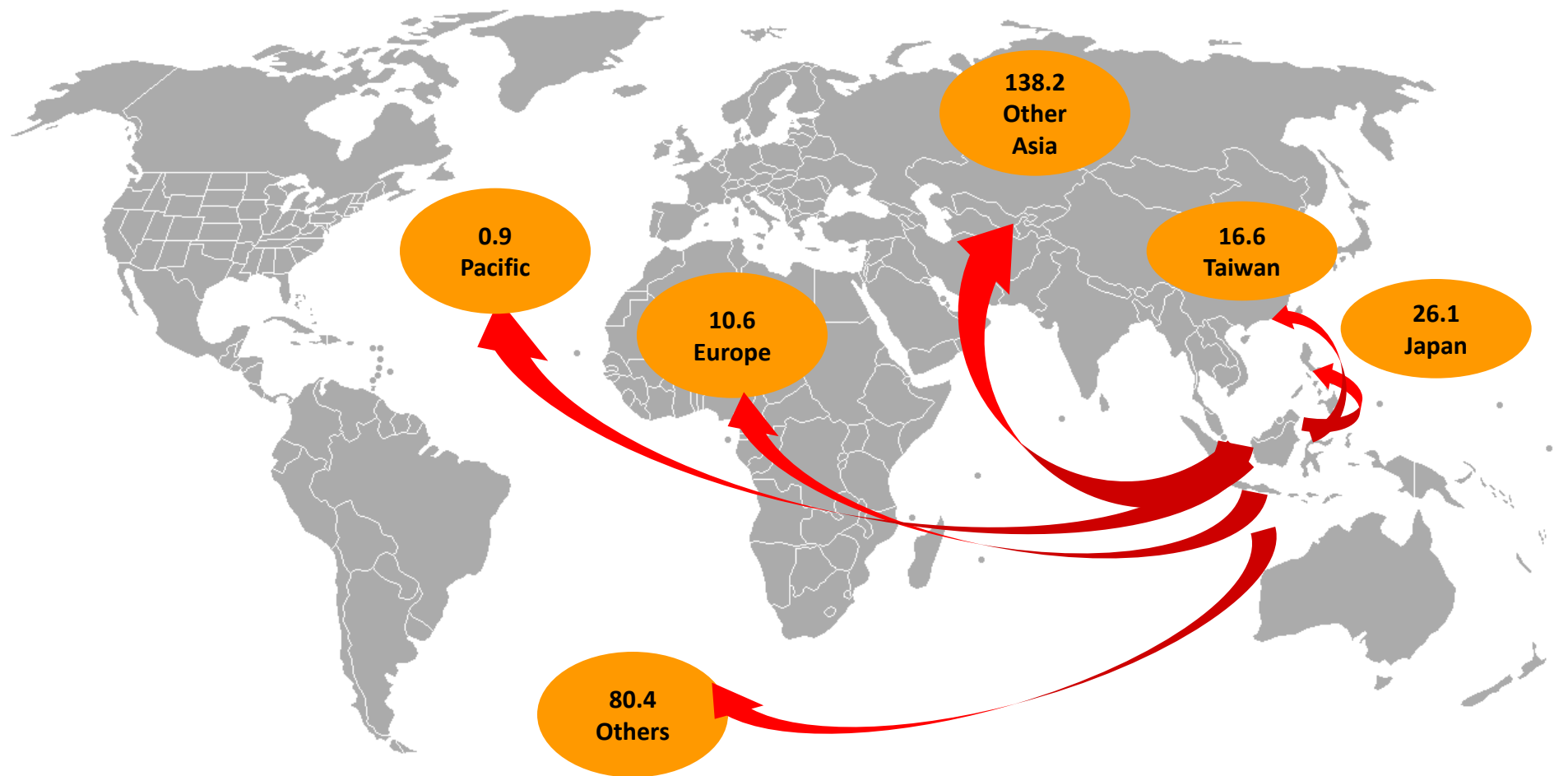
PRODUCTION, EXPORT, DOMESTIC SALE, AND COAL DMO



NOTE : *) Realization forecast in year 2012
Coal DMO started to be enforced in 2010

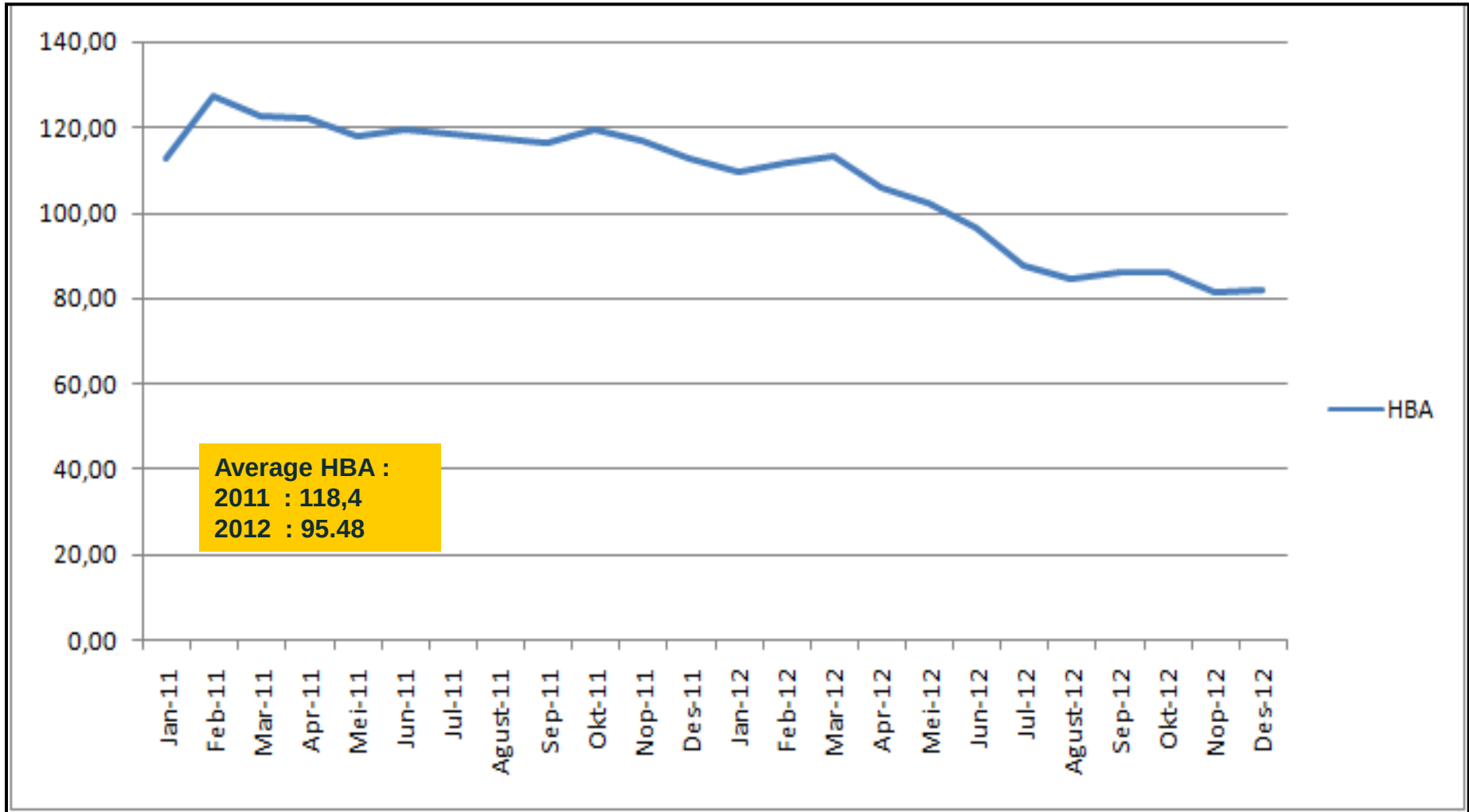
COAL EXPORT BY COUNTRY DESTINATION

(Million Ton)

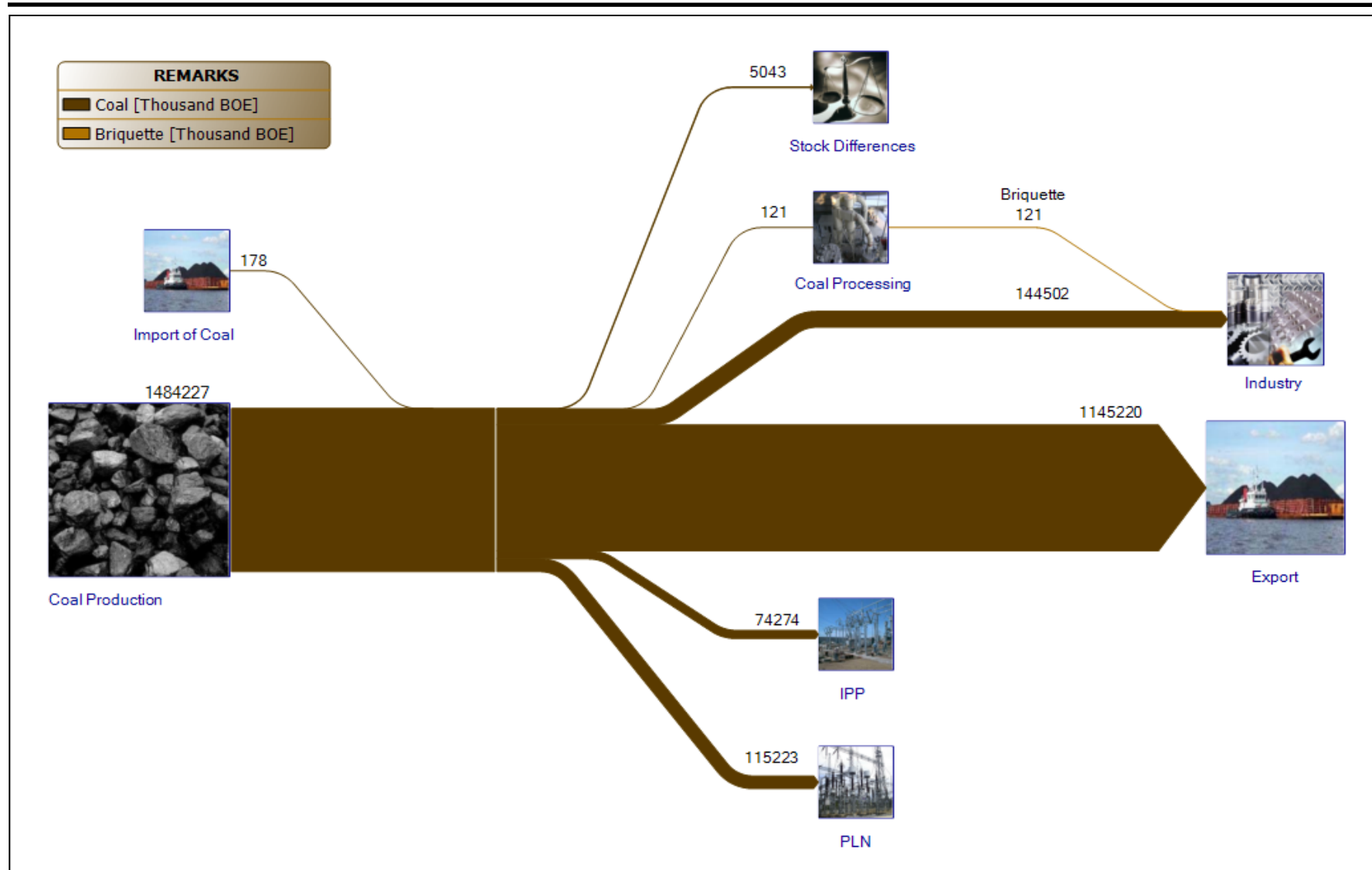


COAL REFERENCE PRICE (HBA)

US Dollar



COAL FLOW 2011



ELECTRICITY

ELECTRIC POWER CONDITION

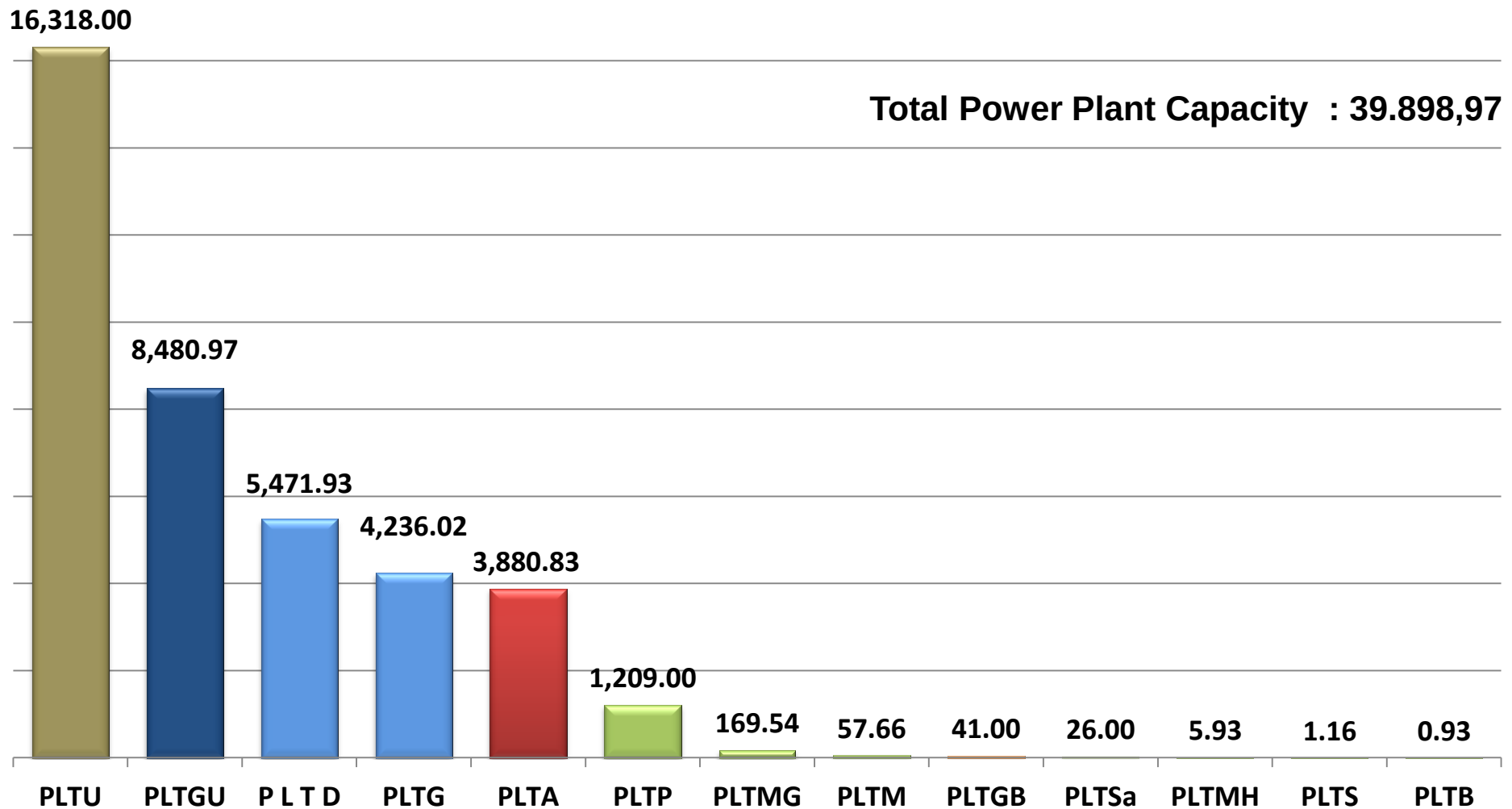
No	DESCRIPTION	UNIT	2011	2012		2013
			Realisation	Plan	Realisation	Plan
1	Demand Growth of Electricity	%	10,1	8,4	8,4	8,6
2	Electrification Ratio	%	72,95	75,3	76,6	79,3
3	Village Electrification Ratio	%	96	96,7	96,7	97,8
4	Total Installed Capacity	MW	39.885	44.224	44.124	48.161
	a. PLN	MW	30.529	33.454	32.108	35.564
	b. <i>Independent Power Producer</i> (IPP)	MW	7.653	9.066	10.287	10.718
	c. <i>Private Power Utilities</i> (PPU)	MW	1.704	1.704	1.729	1.729
5	PLN Electricity Production and Purchase of Electricity	GWh	175.213	190.940	193.663	207.409
6	Rural Electricity					
	a. Sub Station Distribution	MVA	368	213,1	249	216,8
	b. Distribution Network	Kms	17.570,7	8.590,3	11.311,5	9.244,3
	c. Cheap and Efficient Electricity	RTS*)	-	83.000	60.702	95.227

Source : MEMR & PLN

*) RTS = Target Households (<450 Watt)

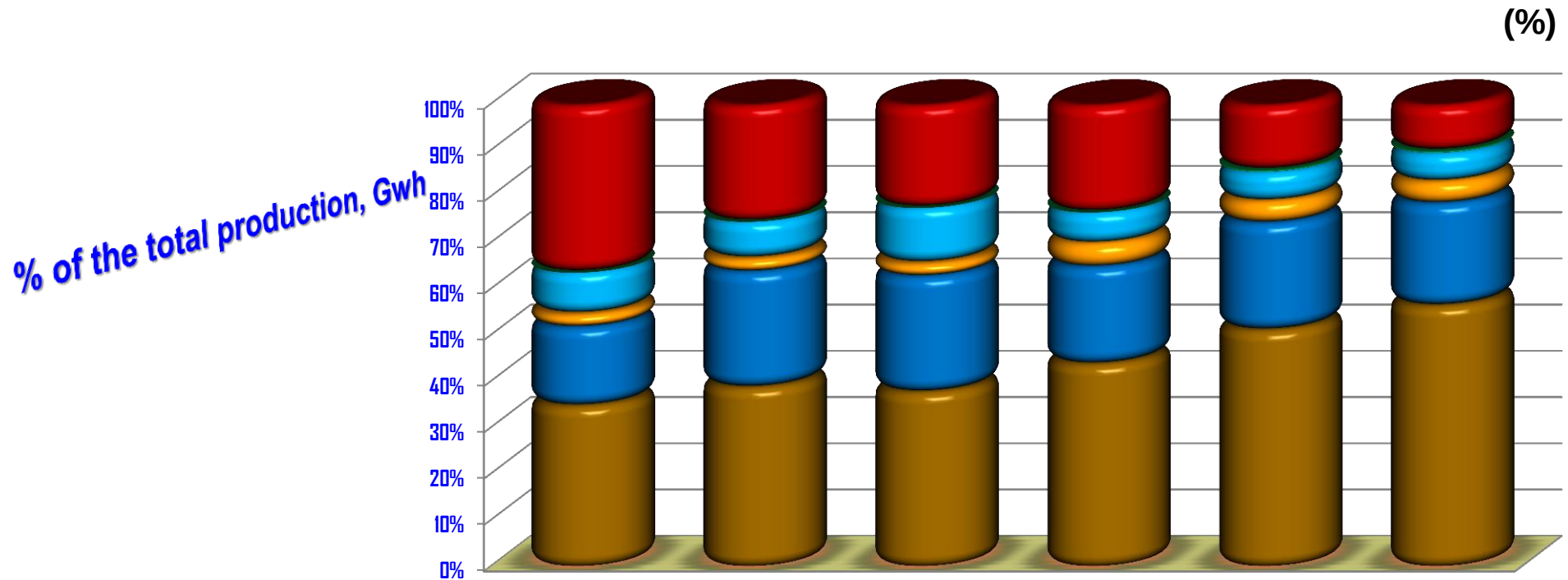
POWER PLANT CAPACITY 2011

(MW)



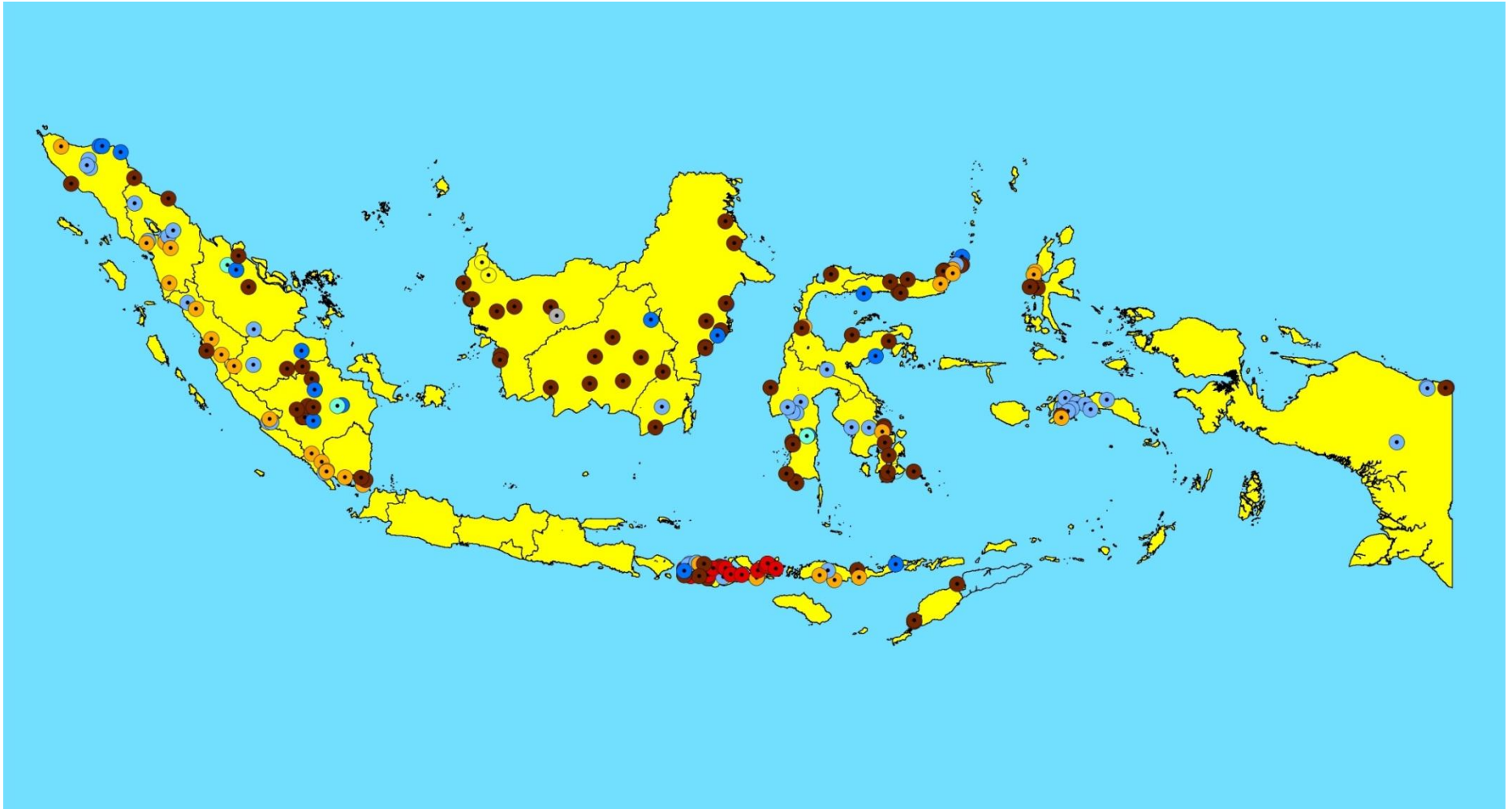
Note : the figures on the PLTU (steamed power plant) does not only use coal as the primary fuel, but there is also the use of gas, the MFO and HSD.

TARGET AND REALISATION OF THE ENERGY MIX POWER PLANT(2008 – 2013)



	2008	2009	2010	2011	2012 *)	2013
 Fuel Oil	36%	25%	22%	22,95%	15,02%	9,70%
 Bio-Diesel & Other NRE	0%	0%	0%	0,07%	0,12%	0,52%
 Hydropower	9%	8%	12%	6,80%	6,44%	6,19%
 Geothermal	3%	3%	3%	5,13%	4,75%	4,80%
 Gas	17%	25%	25%	21,00%	22,60%	22,12%
 Coal	35%	39%	38%	44,06%	51,08%	56,66%

POWER PLANTS SITES PLAN IN INDONESIA



Type of PowerPlant

- | | | | |
|-------------|-------------|-------------|-------------|
| ● PLTA Plan | ● PLGB Plan | ● PLM Plan | ● PLTU Plan |
| ● PLTB Plan | ● PLGU Plan | ● PLTP Plan | ● PLTD Plan |

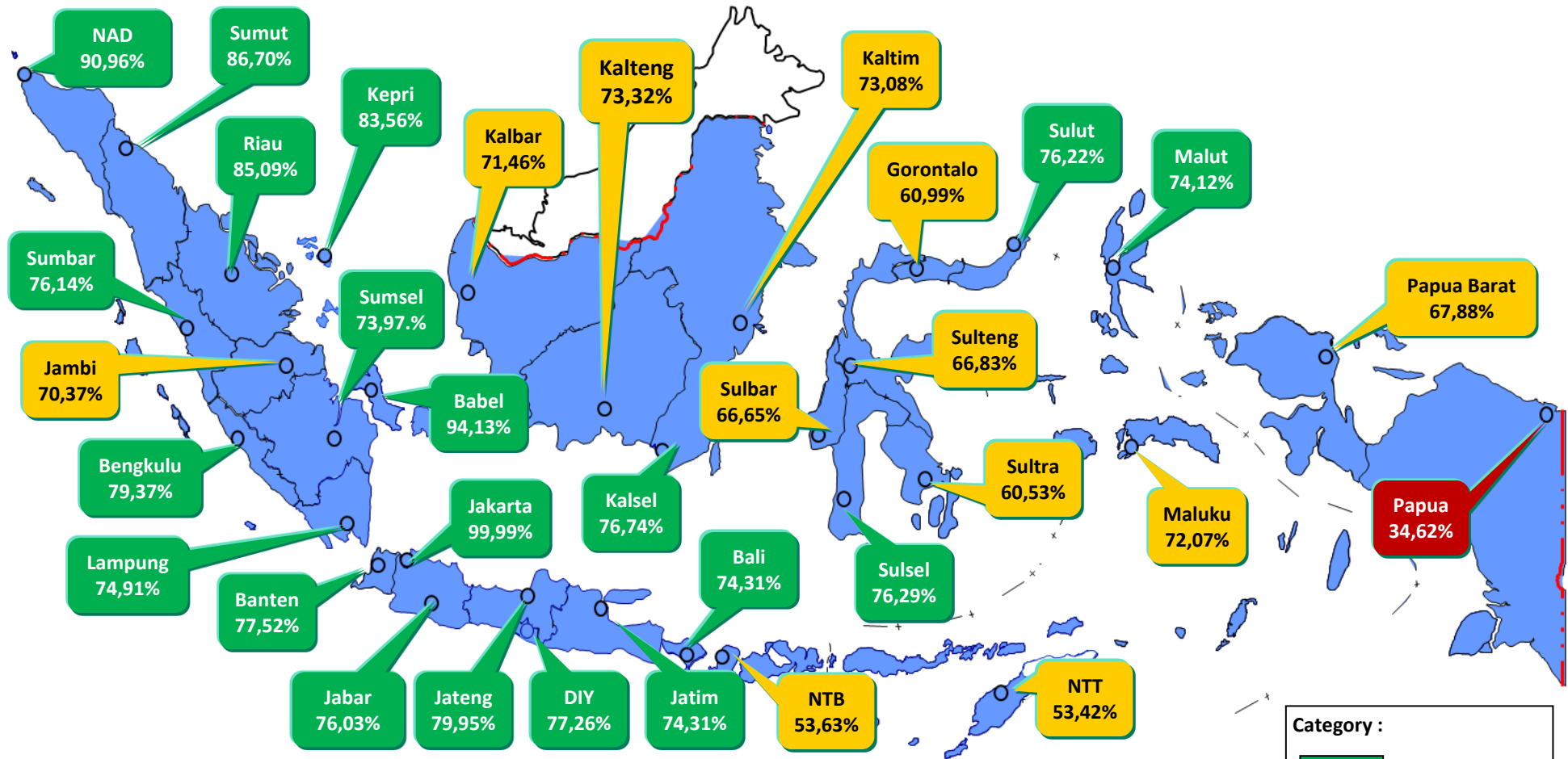
ELECTRICITY TRANSMISSION NETWORK IN INDONESIA



— Existing Transmission Line

..... Transmission Line Plan

ELECTRIFICATION RATIO (2012)



	REALISATION (%)						PLAN (%)	
	2007	2008	2009	2010	2011	2012	2013	2014
Electrification Ratio	64,3%	65,1%	65,8%	67,2%	73,0%	76,6%	79,3%	81,4%
RPJM Target				67,2%	70,4%	73,6%	76,8%	80,0%

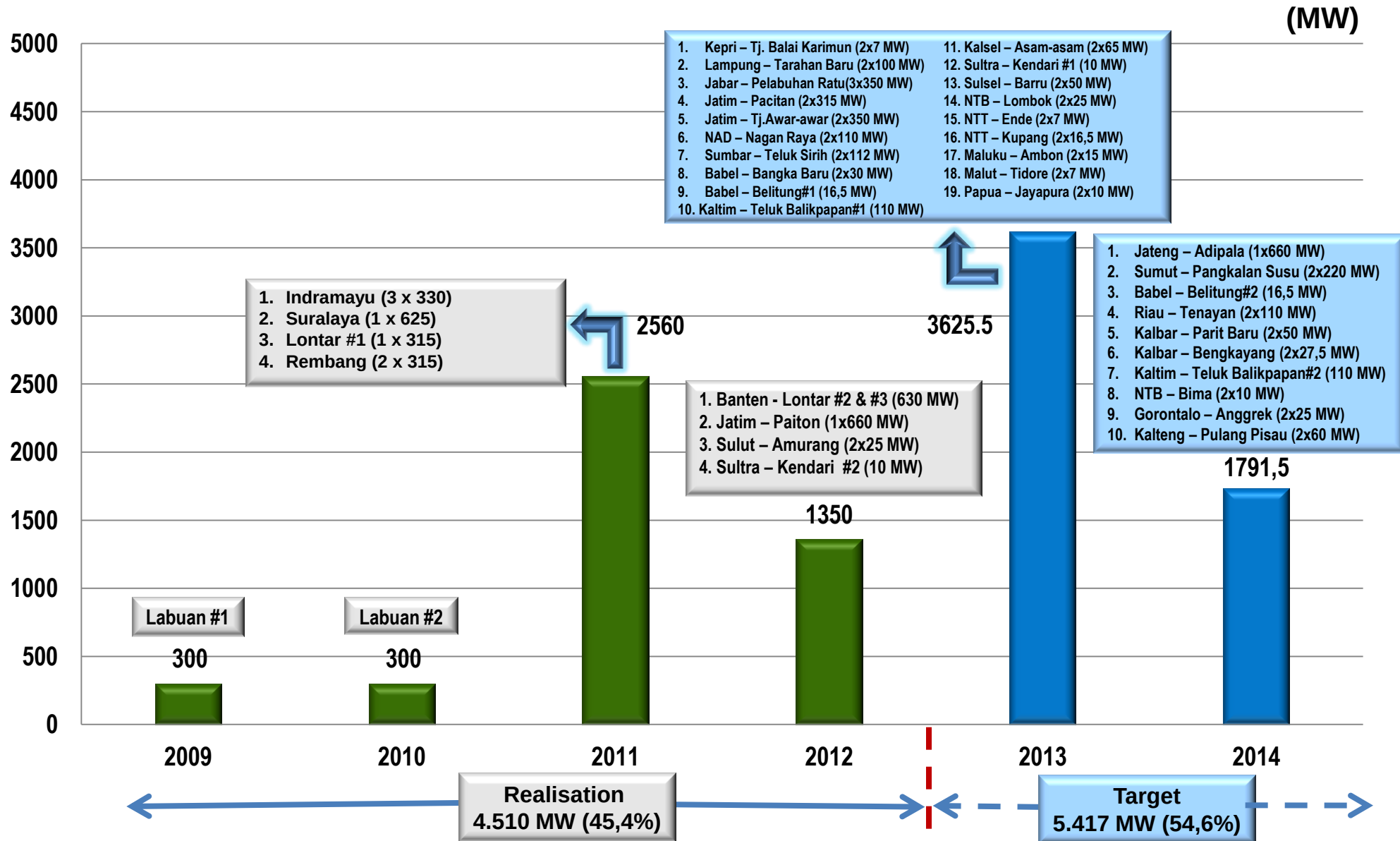
Category :

- > 73,6 %
- 50 – 73,6 %
- 20 - 50 %

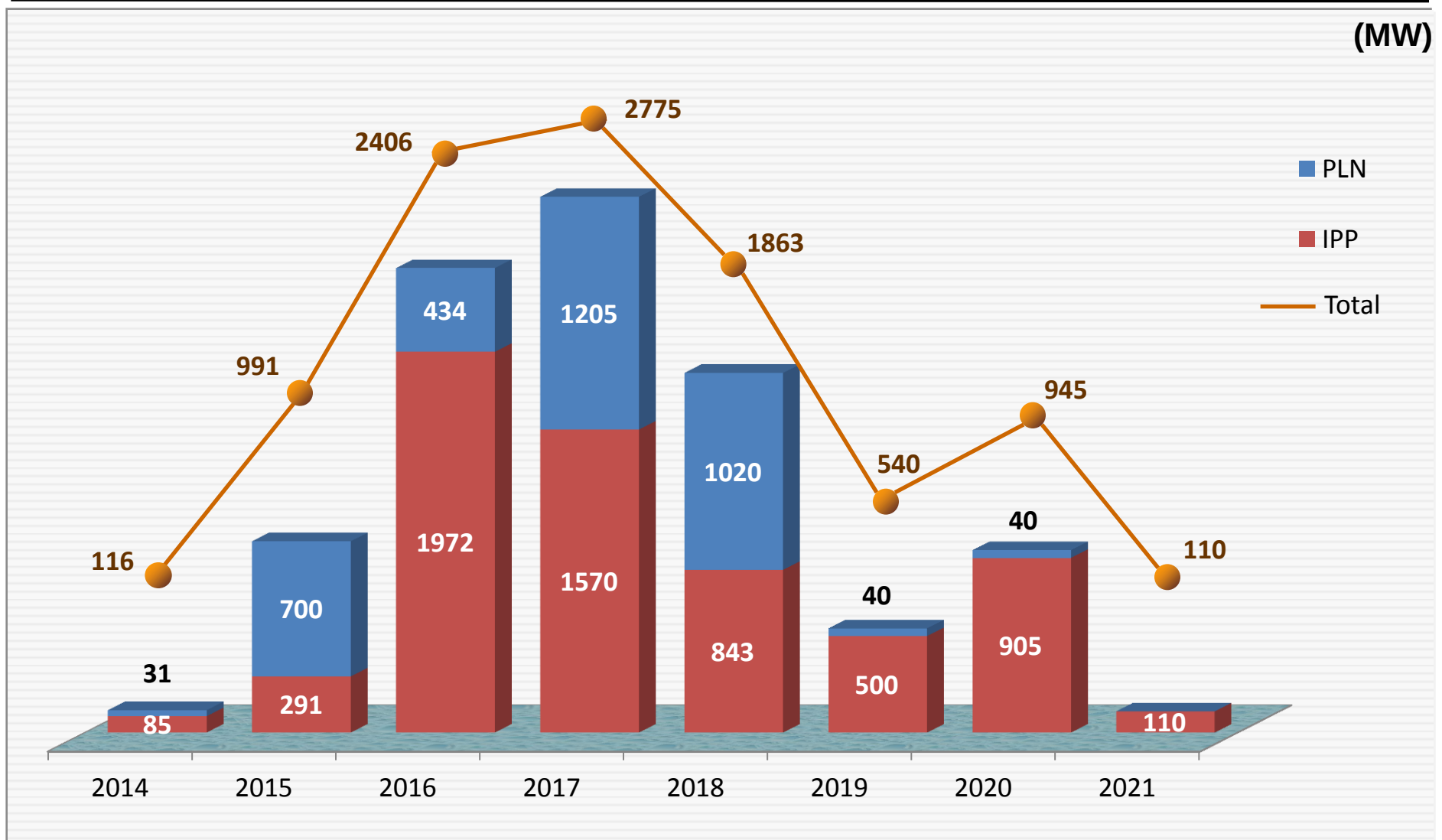
Note : The category to be attuned to the RPJM Target 2012 (C%) ; RPJM = Long and MediumTerm Development Plan

PROGRESS OF FTP-I AND FTP-II

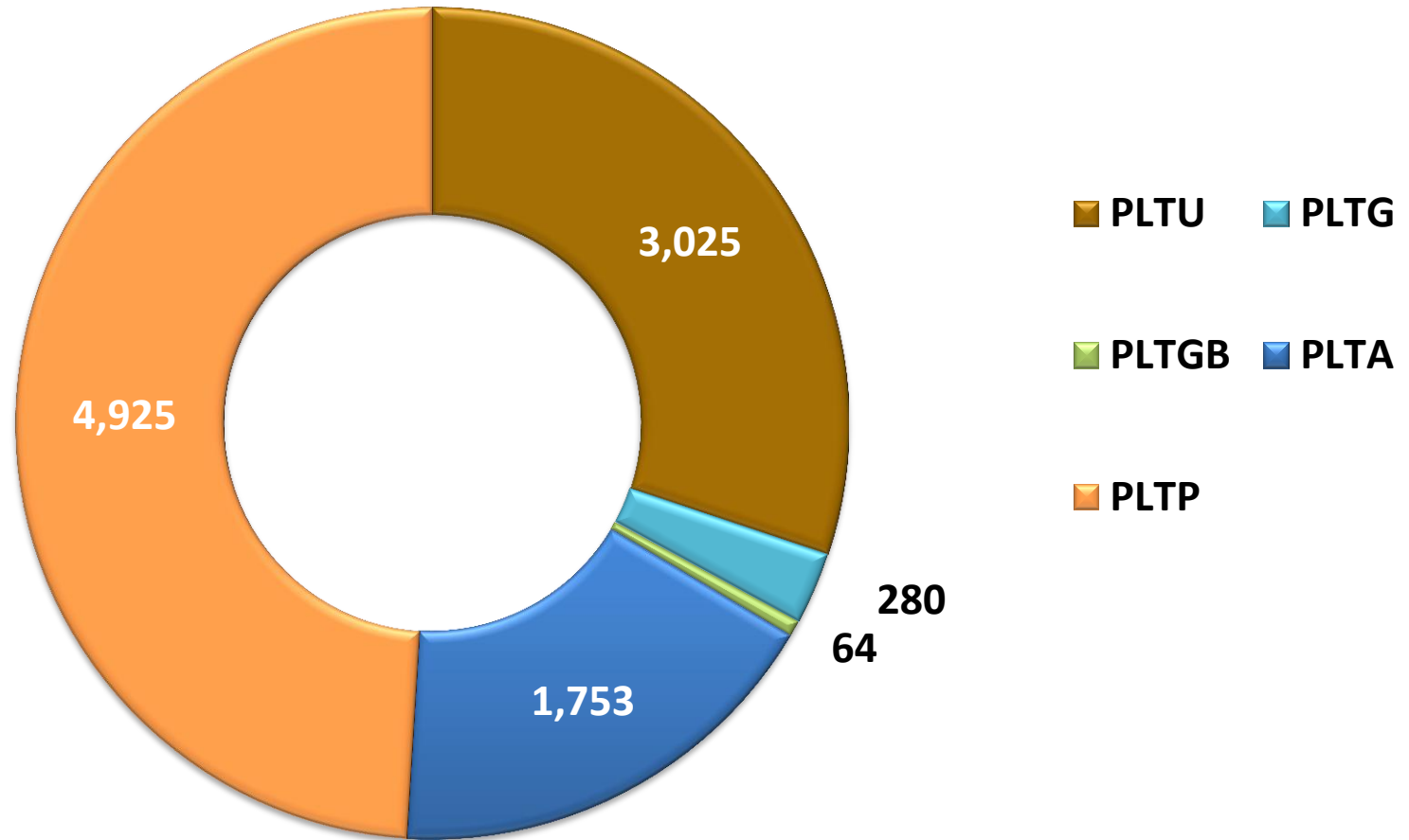
10,000 MW FAST TRACK PROGRAM PHASE I (FTP I)



10,000 MW FAST TRACK PROGRAM PHASE II (FTP II)

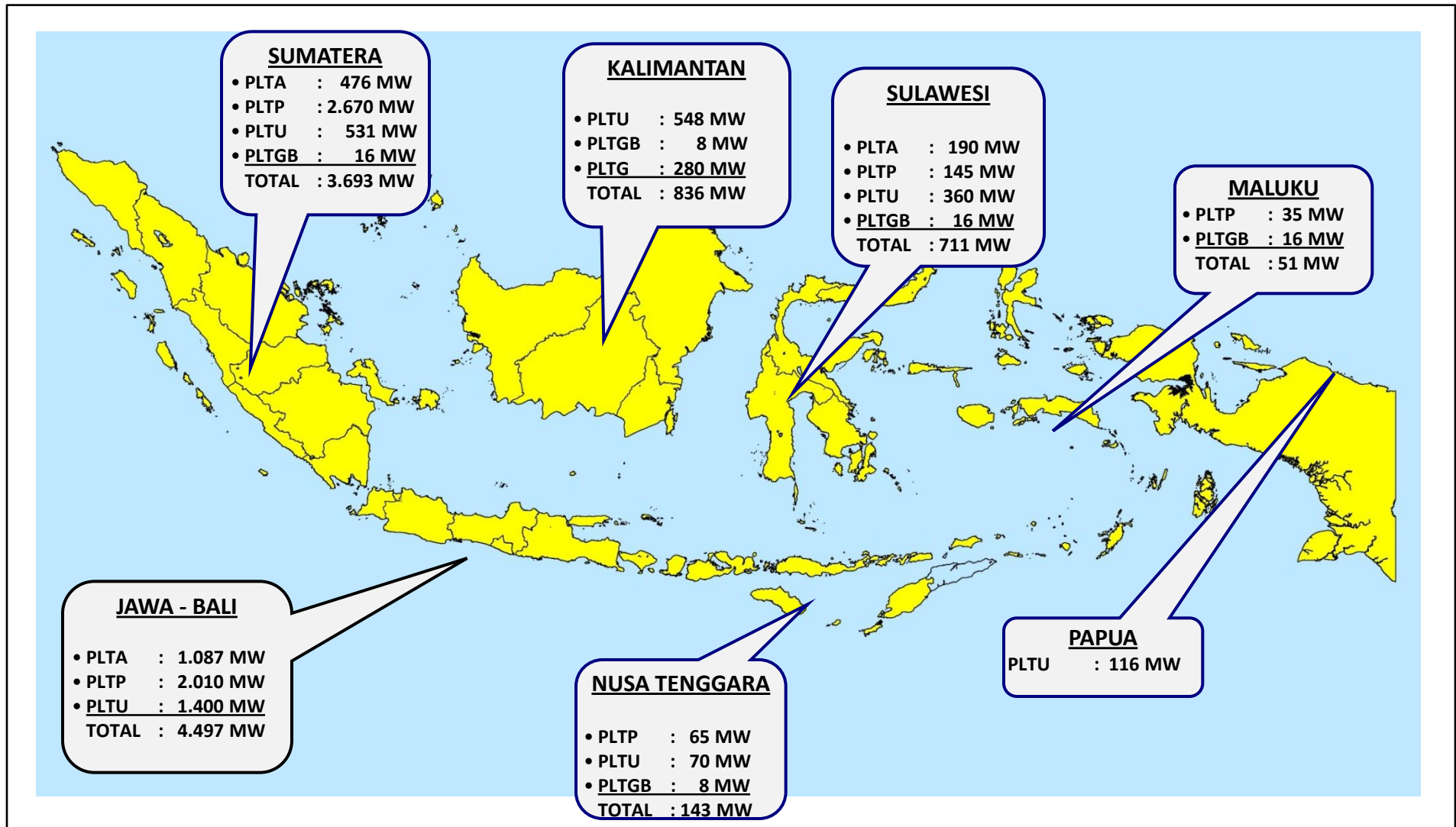


COMPOSITION OF 10,000 MW FAST TRACK PROGRAM PHASE II



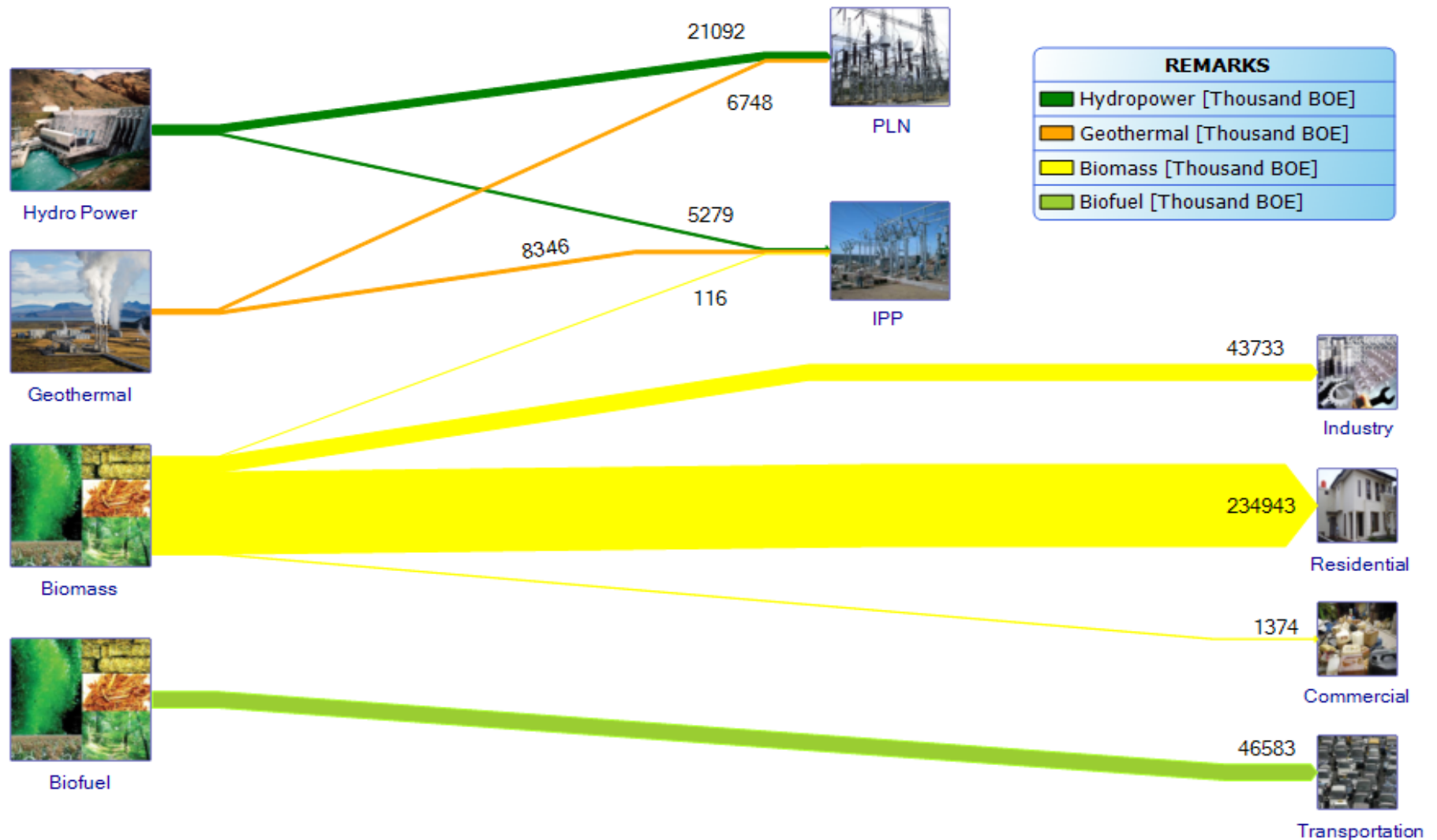
POWER PLANT DISTRIBUTION OF 10,000 MW PROGRAM PHASE II

(EMR Ministerial Regulation Number 01 Year 2012)



NEW AND RENEWABLE ENERGY

NEW AND RENEWABLE ENERGY FLOW 2011



NEW AND RENEWABLE ENERGY RESOURCES

NO	TYPE	RESOURCE	INSTALLED CAPACITT (MW)	RATIO (%)
1	2	3	4	5 = 4/3
1	Hydro (MW)	75.000 MW	6.848,46 MW	9,13%
2	Geothermal (MW)	29.164 MW	1.341 MW	4,6 %
3	Biomass (MW)	49.810 MW	1.644,1 MW	3,3%
4	Solar Energy	4,80 kWh/m ² /day	22,45 MW	-
5	Wind Energy	3 – 6 m/s	1,87 MW	-
6	Ocean	49 GW ²⁾	0,01 MW ³⁾	0%
7	Uranium (MW)	3.000 MW	30 MW ¹⁾	0%

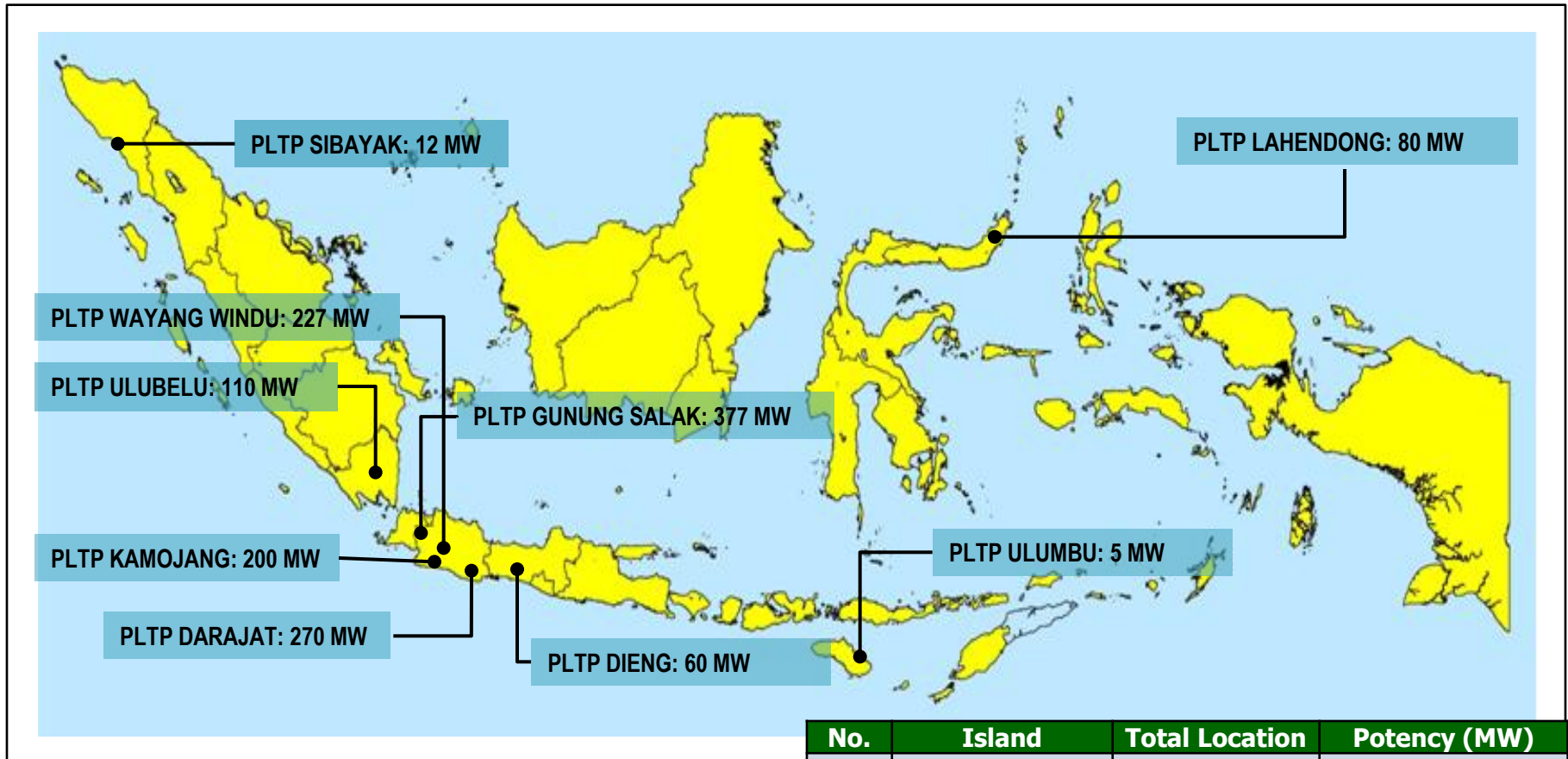
Note:

1) Research scale, non-energy

2) Source: National Energy Council

3) Research Scale: BPPT

GEOHERMAL POTENTIAL



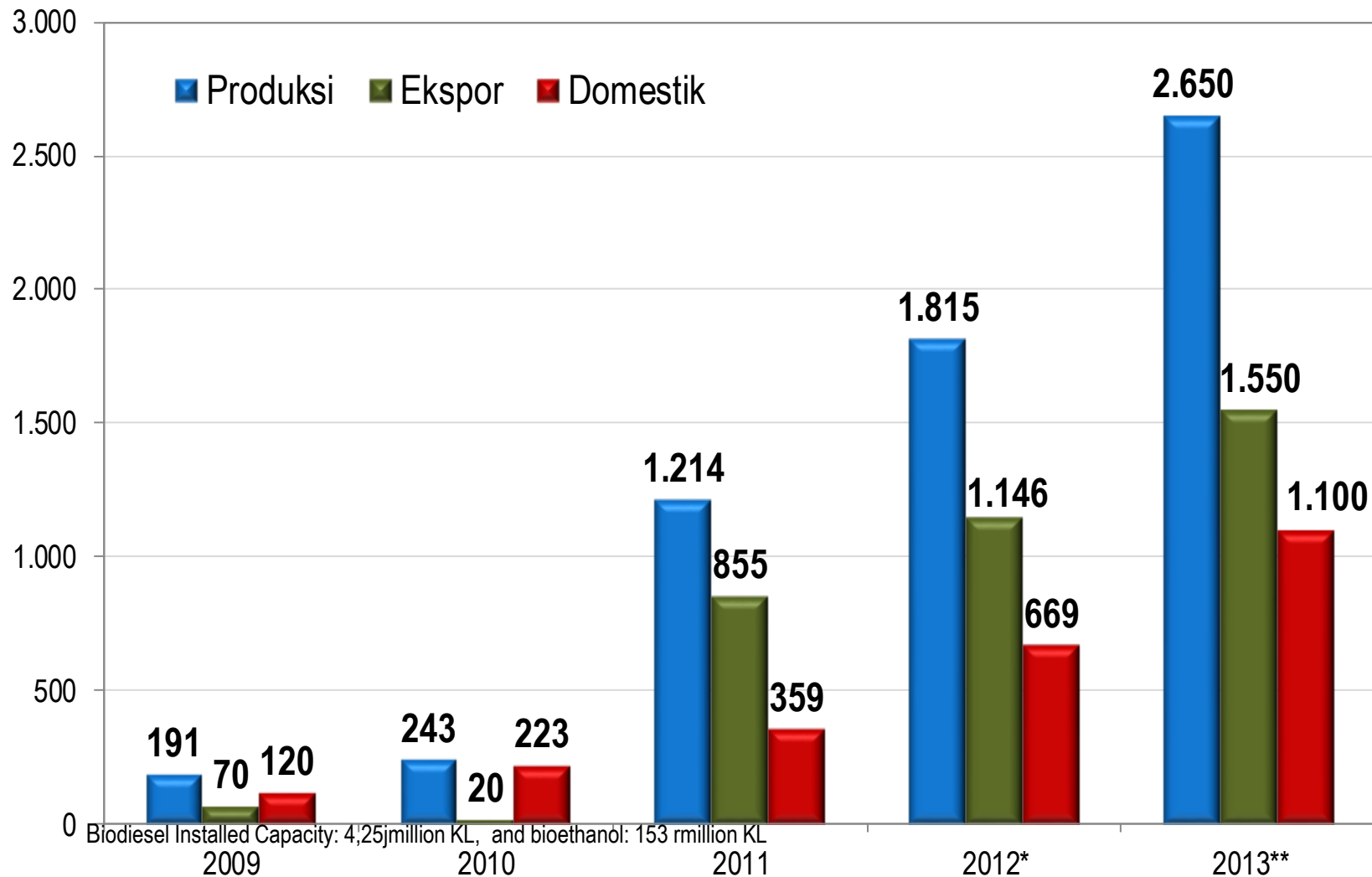
Sumber: Badan Geologi KESDM (2011)

- Preliminary Survey
- Ready to be developed
- Detailed Survey
- Installed

No.	Island	Total Location	Potency (MW)
1	Sumatera	86	13.470
2	Jawa & Bali	76	10.013
3	Nusa Tenggara	22	1.471
4	Kalimantan	12	145
5	Sulawesi	56	2.939
6	Maluku & Papua	33	1.126
TOTAL		285	29.164

NATIONAL BIO-FUEL PRODUCTION (2009 – 2013)

(Thousand KL)



Note : *) tentative figure

**) plan

UTILIZATION OF BIO-FUEL REALISATION VS MANDATORY

Biodiesel

(Juta Liter)

	2009	2010	2011	2012 *	2013
Mandatory	775,9	1.076,1	1.297,0	1.641,0	2.017,0
Realisation	119,4 (15,38 %)	223,1 (20,73 %)	358,8 (27,66 %)	669,4 (40,79 %)	

Bioethanol**

(Juta Liter)

	2009	2010	2011	2012	2013
Mandatory	215,8	660,9	694,0	968,0	1.167,0
Realisation	1.058 (0,49 %)	-	-	-	

* Percentage of Biodiesel blending with diesel oil at 7.5% as of February 15, 2013

** There has been no actual utilization of Bioethanol as the revised Market Price Index has not yet been approved

*** Bio-fuel by the Ministry of Finance

CO2 Emissions

NATIONAL COMMITMENT TO REDUCE GHG

President Commitment in G-20 Pittsburgh and COP15 to reduce greenhouse gasses (GHG) emission by 2020

Domestic effort

26%
(767 million ton)

Domestic effort and international support

41%
(1209 million ton)

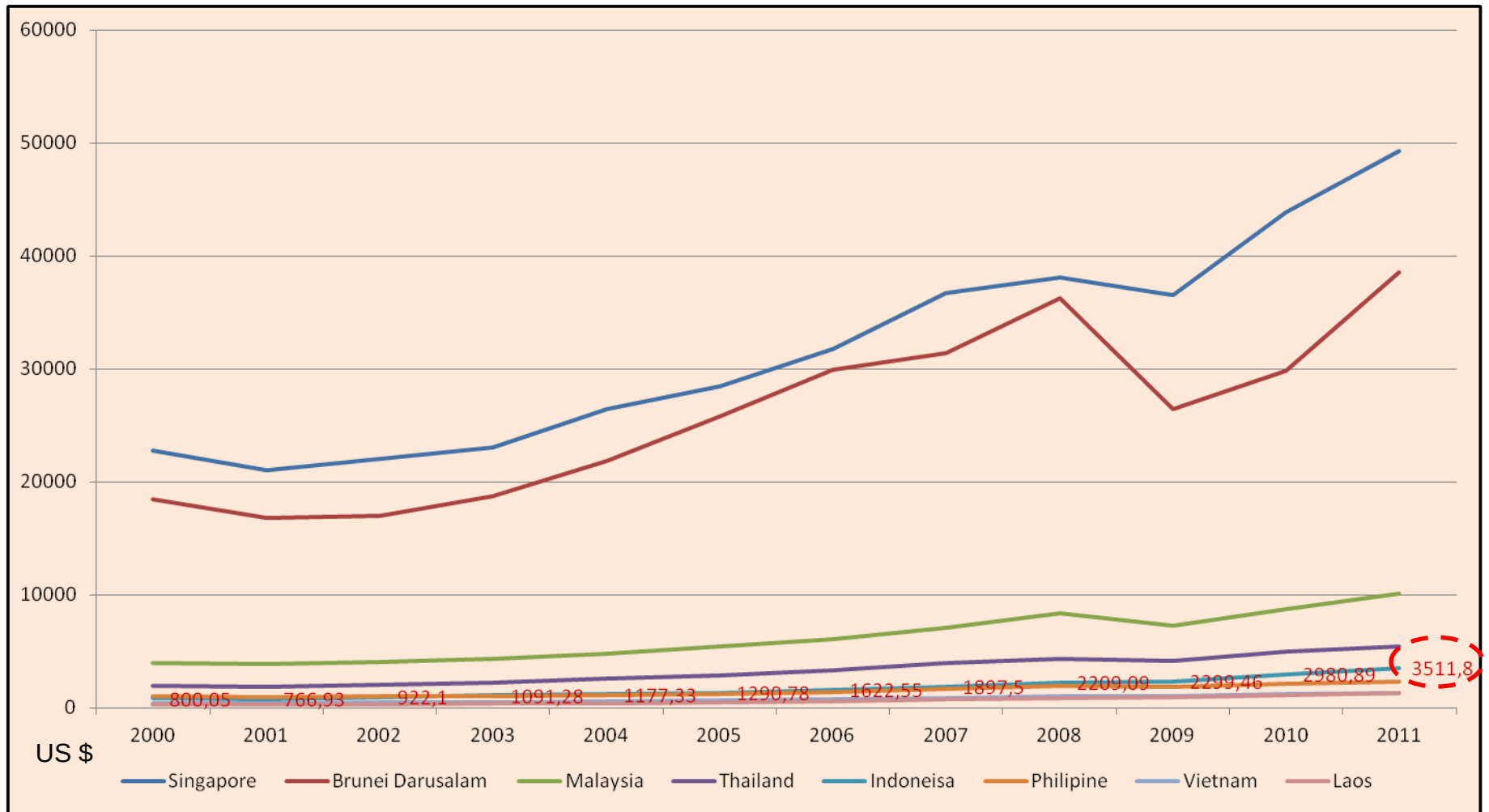
New Energy Development

Energy Conservation

Forestry, Peat, Agriculture	680 million Ton
Energy Sector	30 million Ton
Wastes	48 million Ton
Industry and Transport	9 Juta Ton

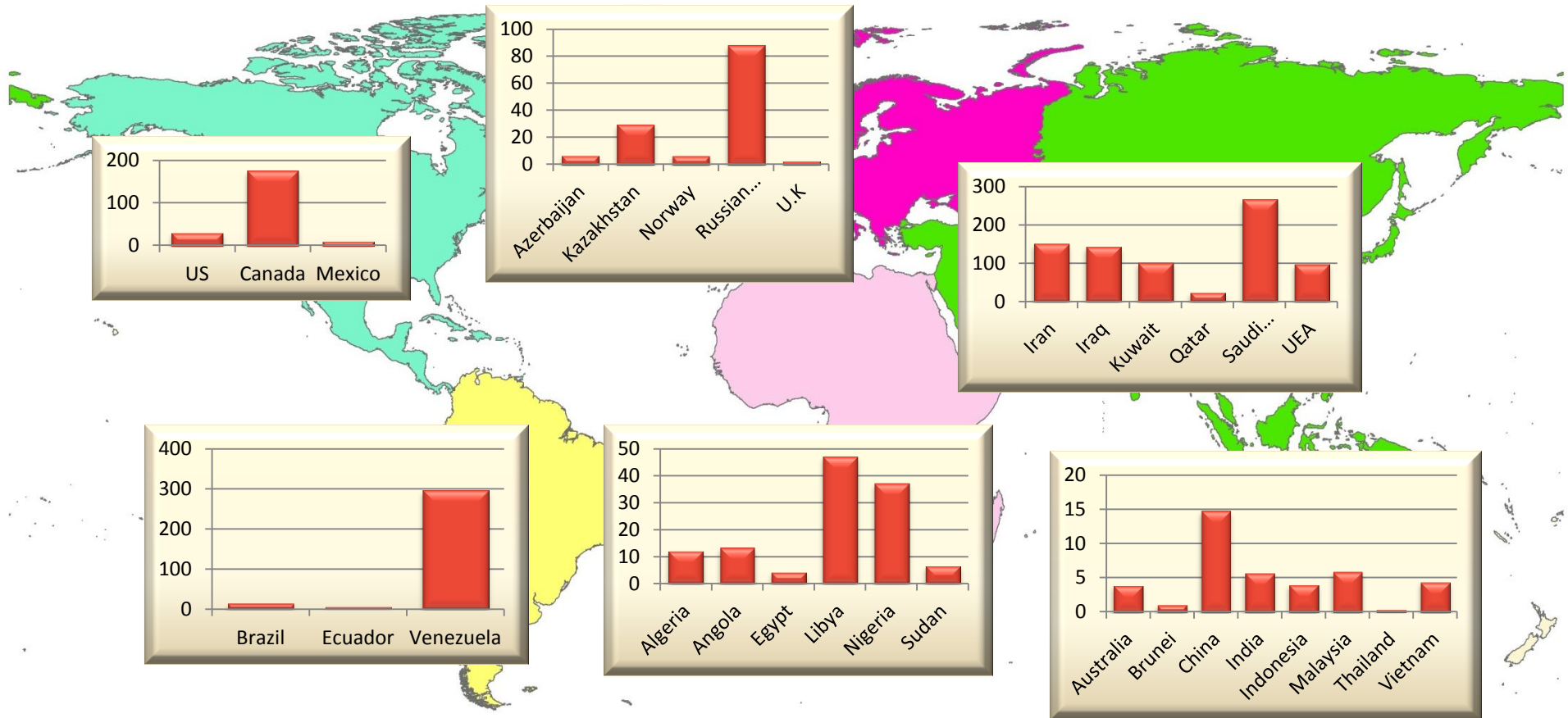
INDONESIA'S POSITION

GDP PER CAPITA IN ASEAN MEMBER COUNTRIES



WORLD OIL RESERVES

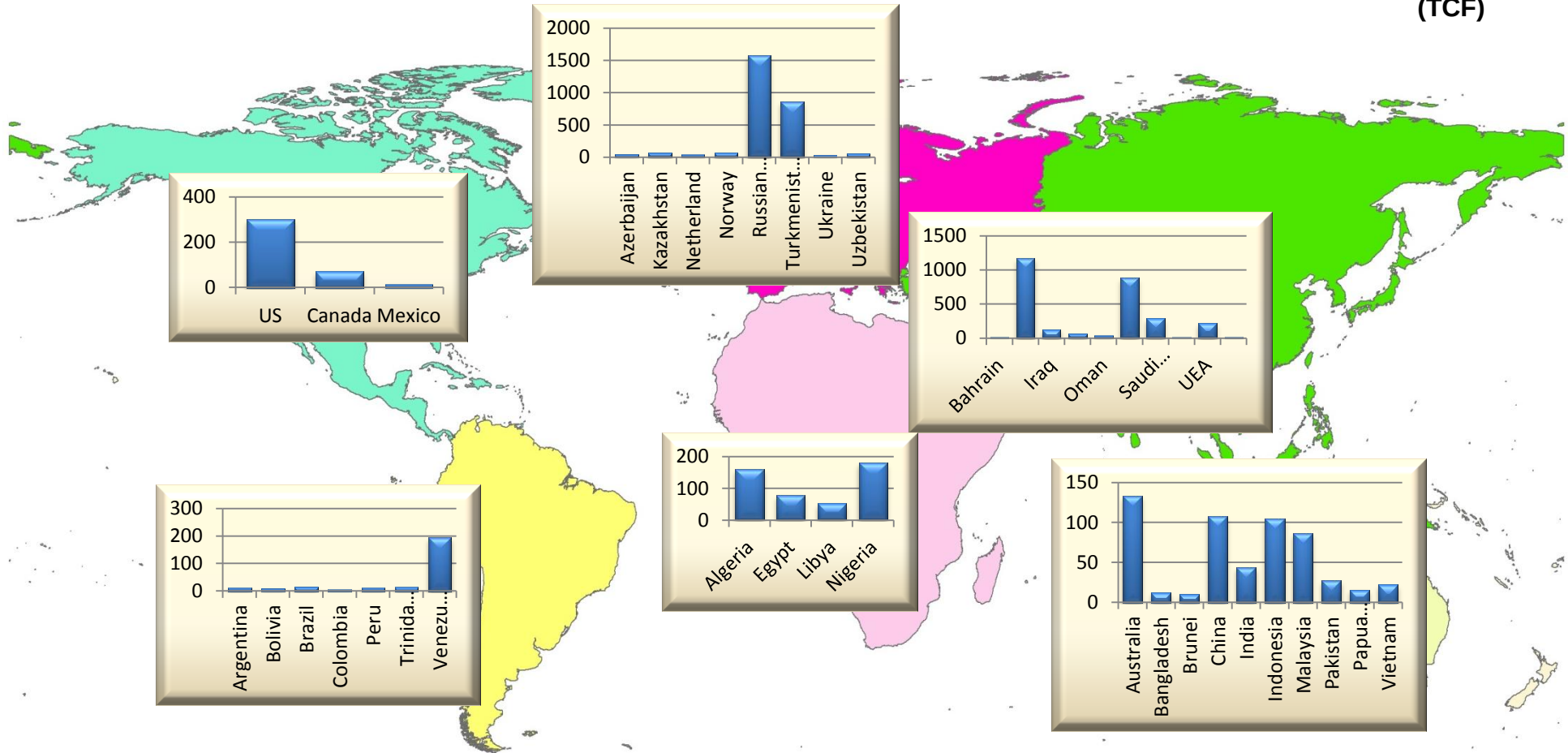
(Thousand MMbbl)



BP Statistical Review, by 2011

WORLD NATURAL GAS RESERVES

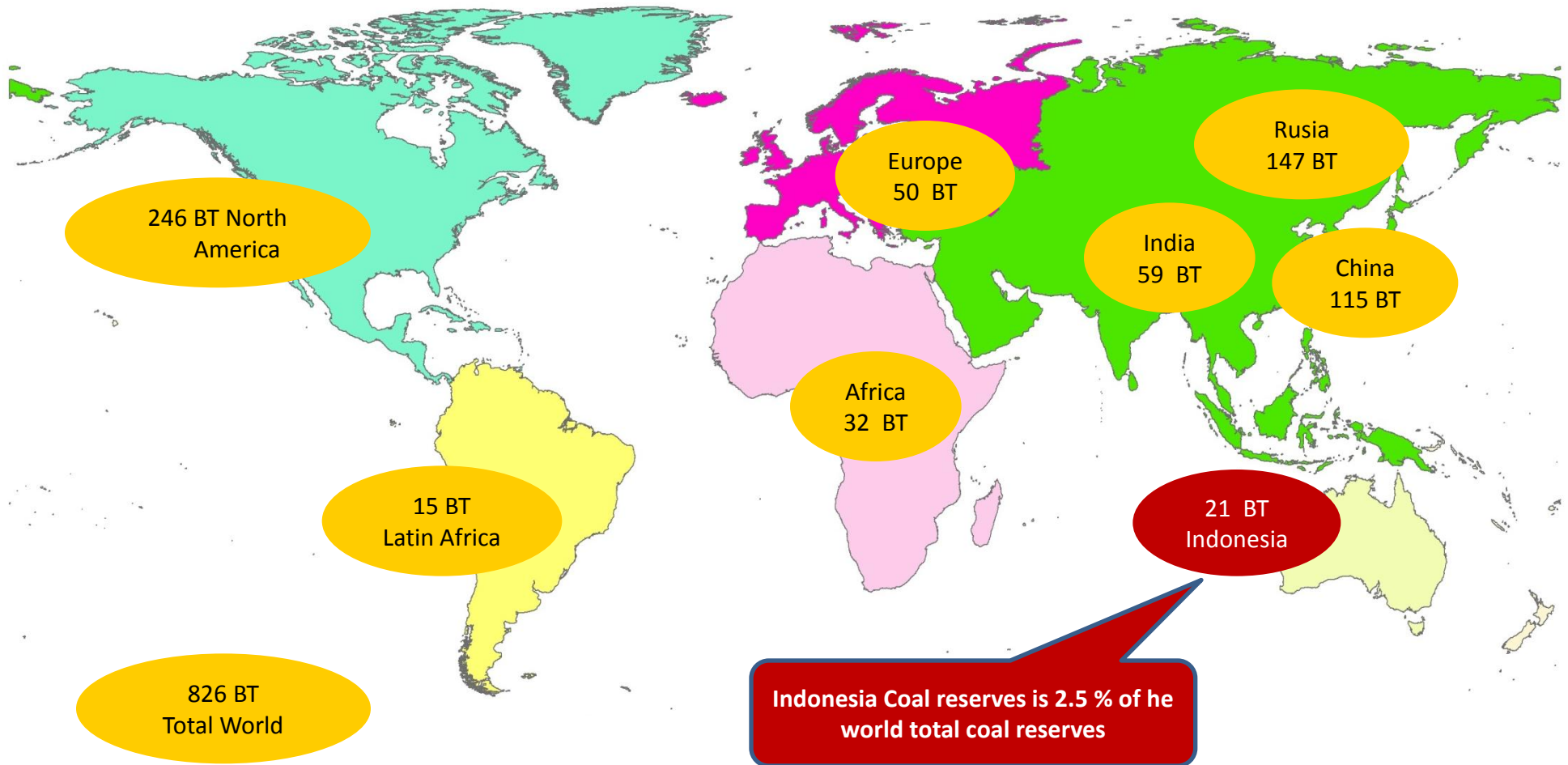
(TCF)



By the end of 2011

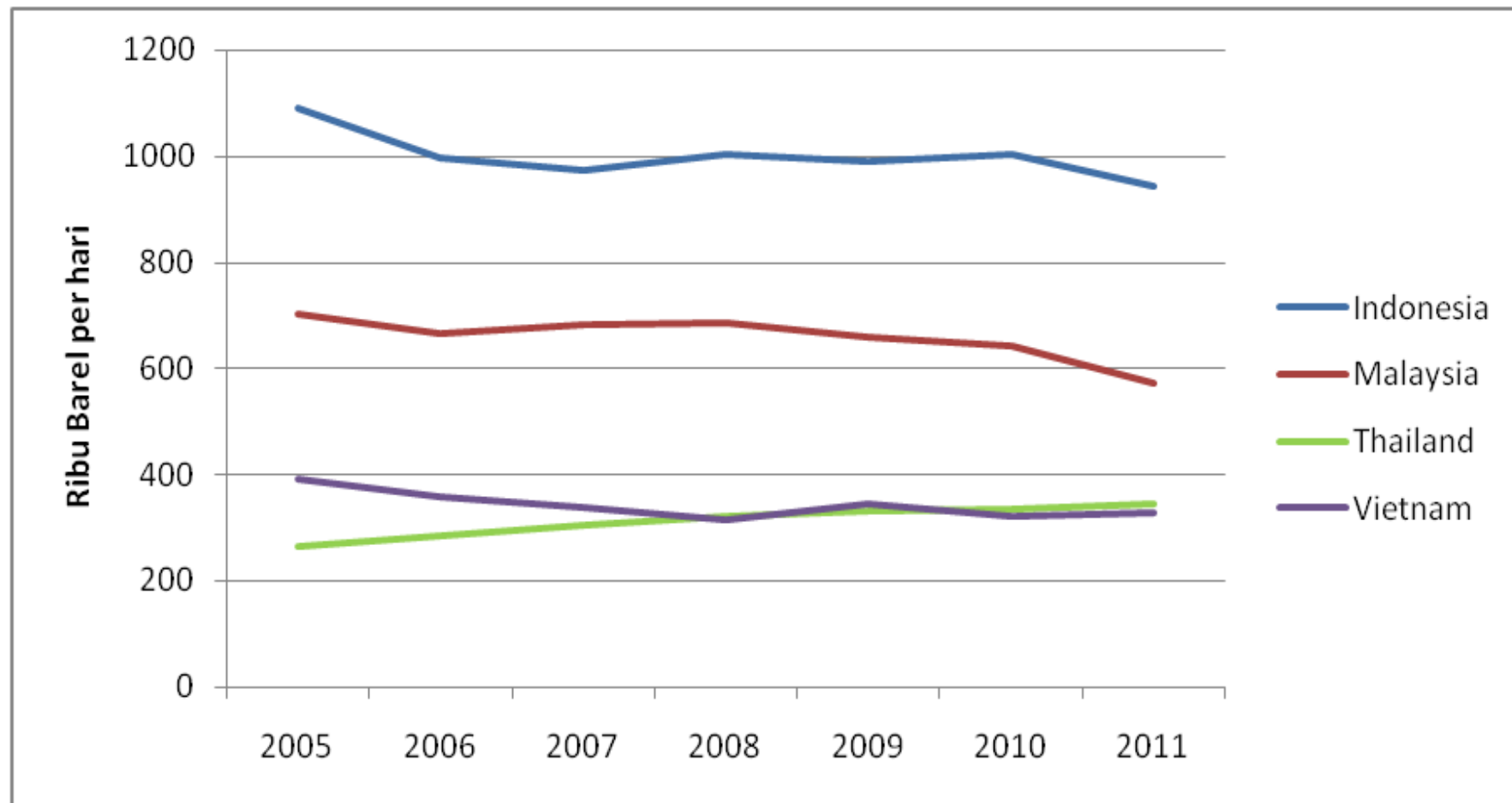
WORLD COAL RESERVES

Billion Ton



ASEAN OIL PRODUCTION

(Thousand Barrel per day)

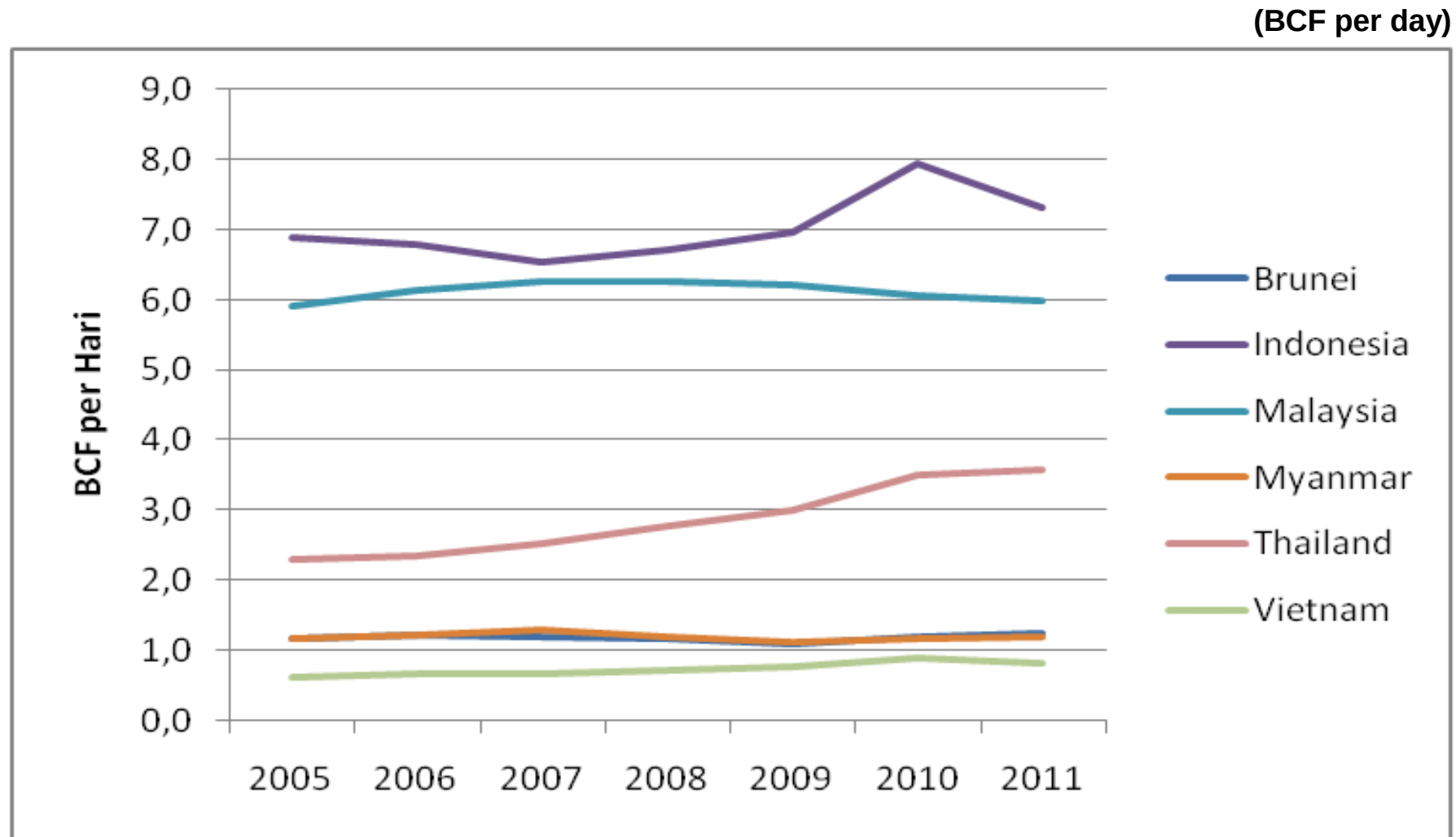


Note:

Data of Other ASEAN member states are not yet provided

Source: BP statistical Review

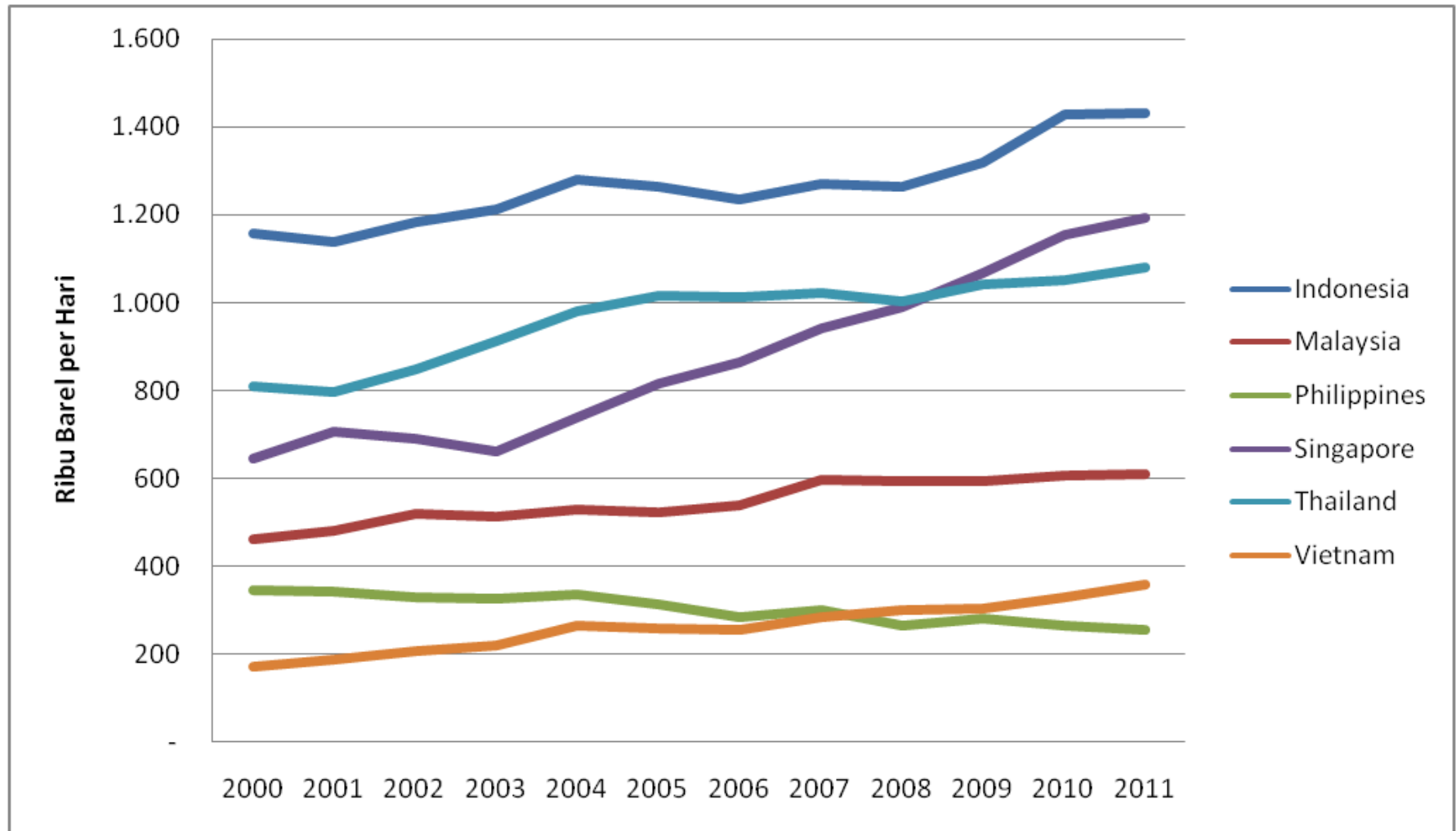
ASEAN GAS PRODUCTION



Note : data of ther ASEAN Member States are not yet provided

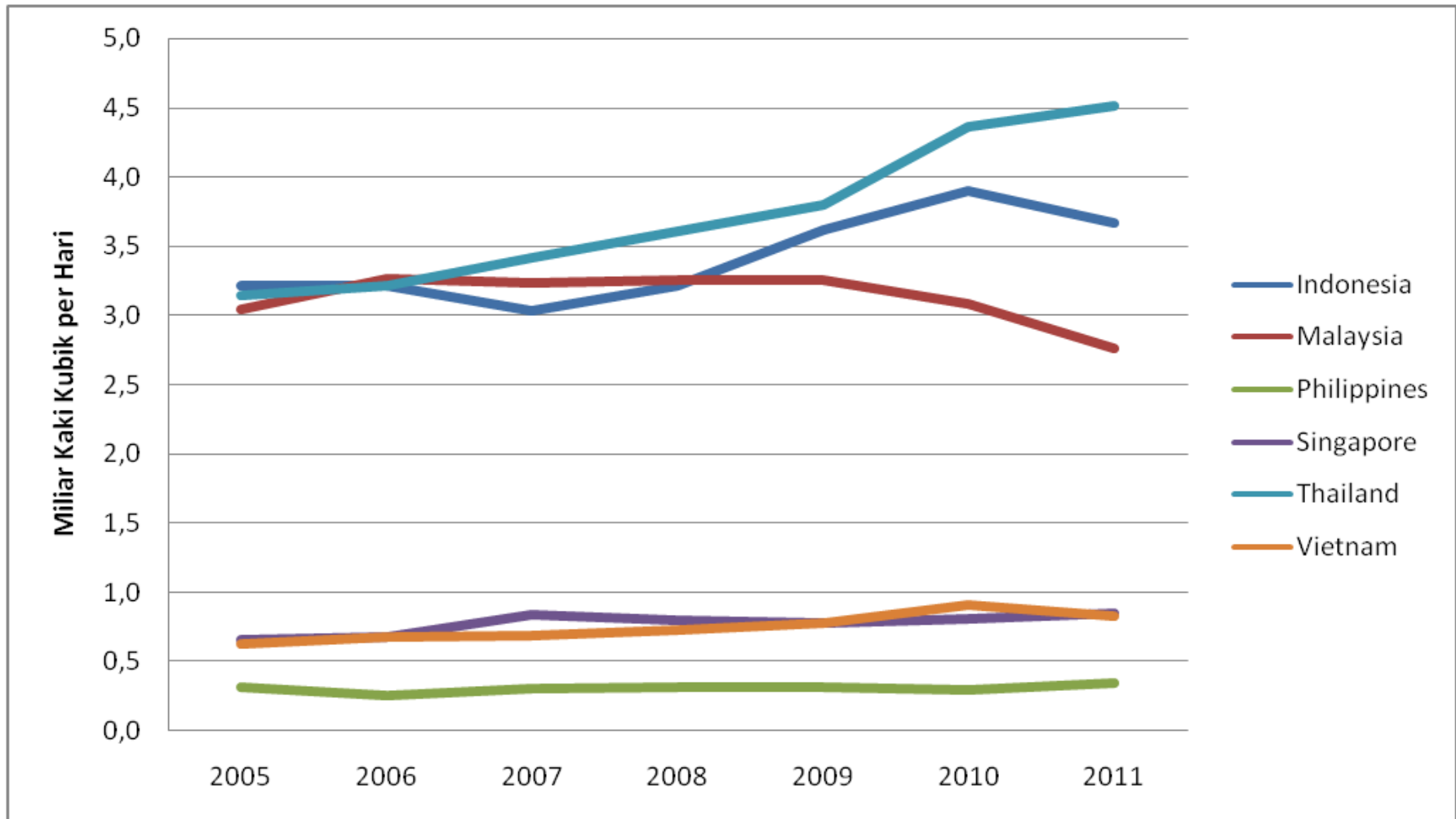
ASEAN OIL CONSUMPTION

(Thousand Barrel per day)



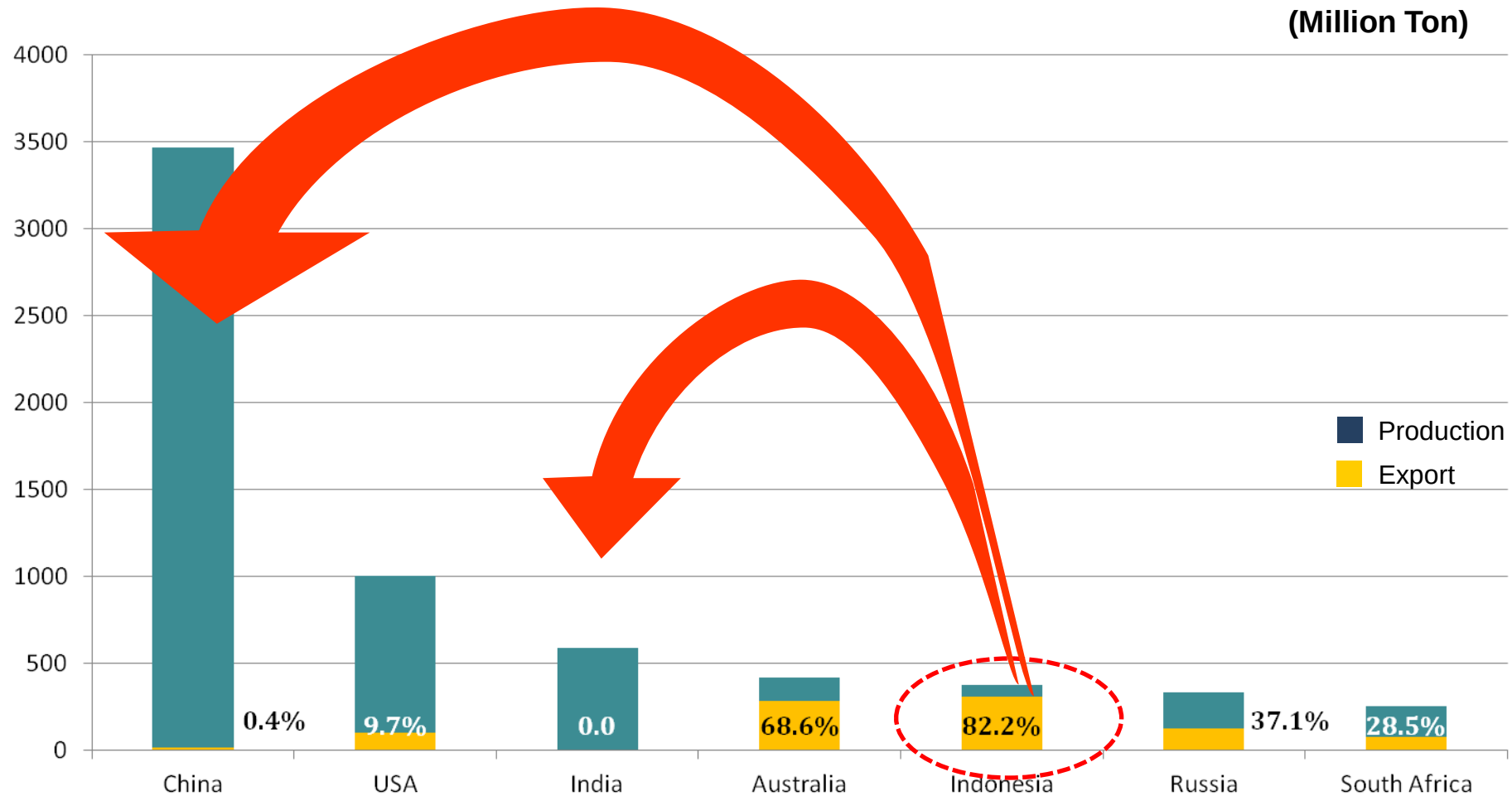
Note: data of other ASEAN Member States are not yet provided.

ASEAN GAS CONSUMPTION



Note : data of other ASEAN Member States are not yet provided

COAL PRODUCTION-EXPORT COMPARISON



Note: In 2011, Indonesian coal exports to China was 92,15 million and to India was 64,371 million .tons

Comparison of Electric Power Consumption in ASEAN Member States

No.	Member state	Population (million people)	Power Plants (Billion kWh)	GDP/ Capita	Power per Capita (kWh/Capita)
1	Philippines	97.98	57	3,100	588
2	Indonesia	240.27	142	3,500	591
3	Malaysia	25.72	103	14,200	3,490
4	Singapore	4.66	41	48,500	8,185
5	Thailand	65.91	148	7,900	2,079
6	Brunei Darussalam	0.39	3	54,400	8,308
7	Vietnam	86.97	61	2,500	799
8	Laos	6.83	2	1,900	178
9	Myanmar	48.14	6	1,100	97
10	Cambodia	14.49	1	1,800	113

Catatan : data terakhir tahun 2010

Thank You

www.den.go.id